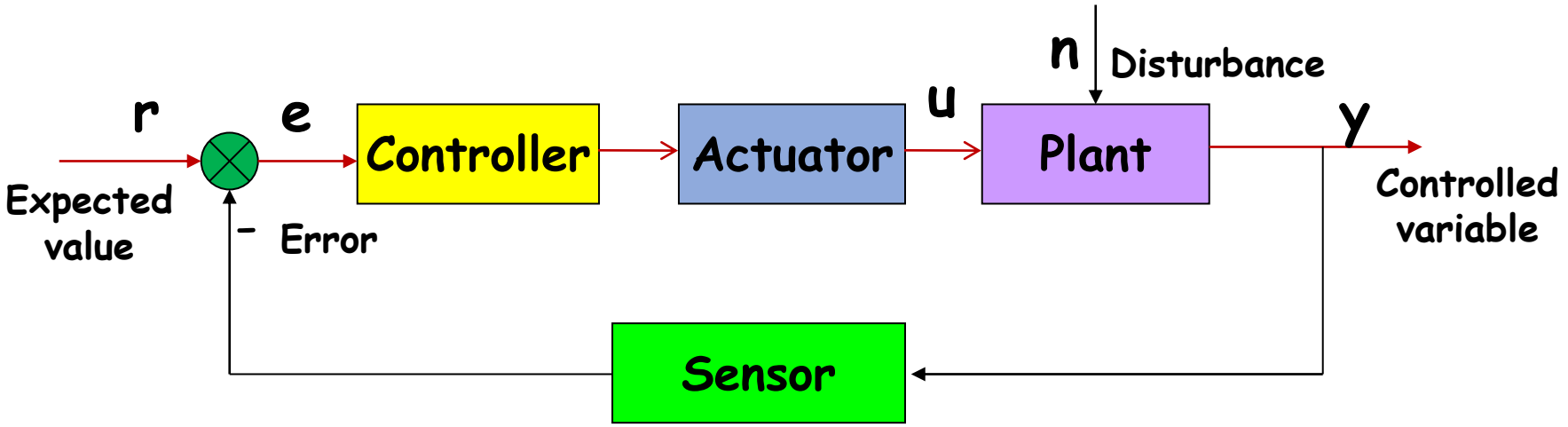
EE 3072
System Dynamics and Control

Lecture 2

Modeling of Control Systems

How to analyze and design a control system



- The first thing is to establish system model (mathematical model)

Outline

- Introduction
- Establishment of differential equation and linearization
- Transfer function
- Structure diagram
- Signal-flow graphs

Introduction on mathematical models

System model

Definition:

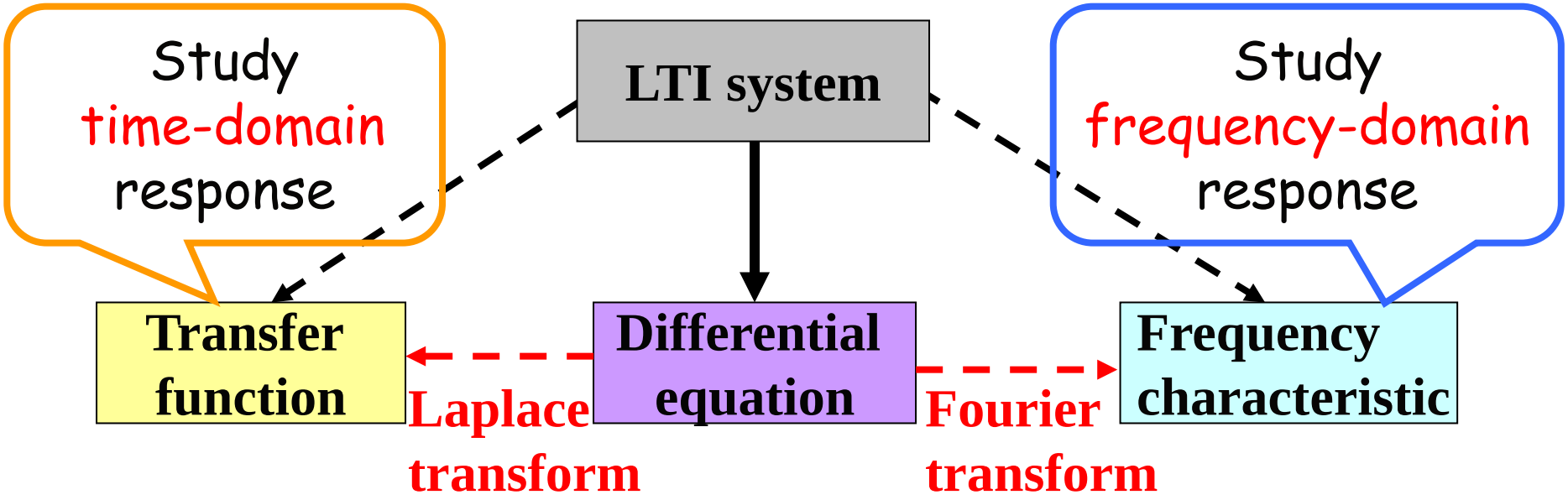
Mathematical expression of dynamic relationship between input and output of a control system.

Mathematical model is foundation to analyze and design automatic control systems

No mathematical model of a physical system is exact. We generally strive to develop a model that is adequate for the problem at hand but without making the model overly complex.

Three models

- Differential equation
- Transfer function
- Frequency characteristic



Modeling methods

Analytic method

According to

A. Newton's Law of Motion
B. Kirchhoff's Laws
C. Laws of Physics
D. System structure and parameters

the mathematical expression of system input and output can be derived.

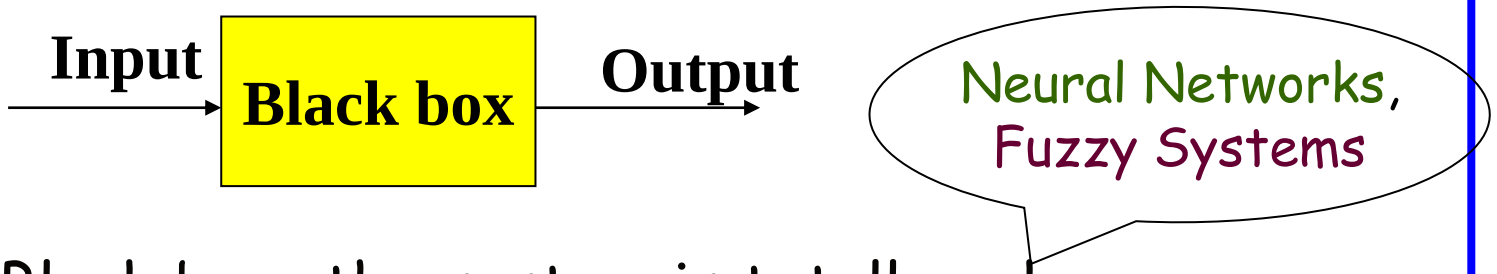
Thus, we build the mathematical model (suitable for simple systems).

Modeling methods

System identification method

Building the system model based on the system input—output signal

This method is usually applied when there is little information available about the system.



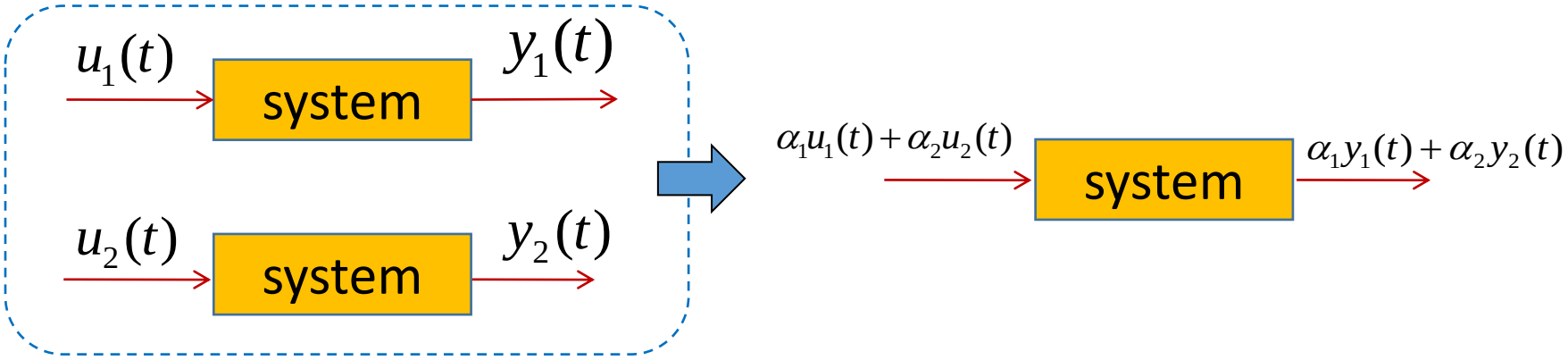
Black box: the system is totally unknown.

Grey box: the system is partially known.

Why Focus on Linear Time-Invariant (LTI) System

- What is linear system?

  -A system is called linear if the principle of superposition applies





Is y(t)=u(t)+2 a linear system?

Advantages of linear systems

- The overall response of a linear system can be obtained by

  -- decomposing the input into a sum of elementary signals

  -- figuring out each response to the respective elementary signal

  -- adding all these responses together.

Thus, we can use typical elementary signal (e.g. unit step, unit impulse, unit ramp) to analyze system for the sake of simplicity.

- What is time-invariant system?
  - A system is called time-invariant if the parameters are stationary with respect to time during system operation
  - Examples?



Establishment of differential equation and linearization

Differential equation

- Linear ordinary differential equations

--- A wide range of systems in engineering are modeled mathematically by differential equations

--- In general, the differential equation of an *n*-th order system is written

$$a_0 c^{(n)}(t) + a_1 c^{(n-1)}(t) + \cdots + a_{n-1} c^{(1)}(t) + a_n c(t) =$$
$$b_0 r^{(m)}(t) + \cdots + b_{m-1} r^{(1)}(t) + b_m r(t)$$

Time-domain model

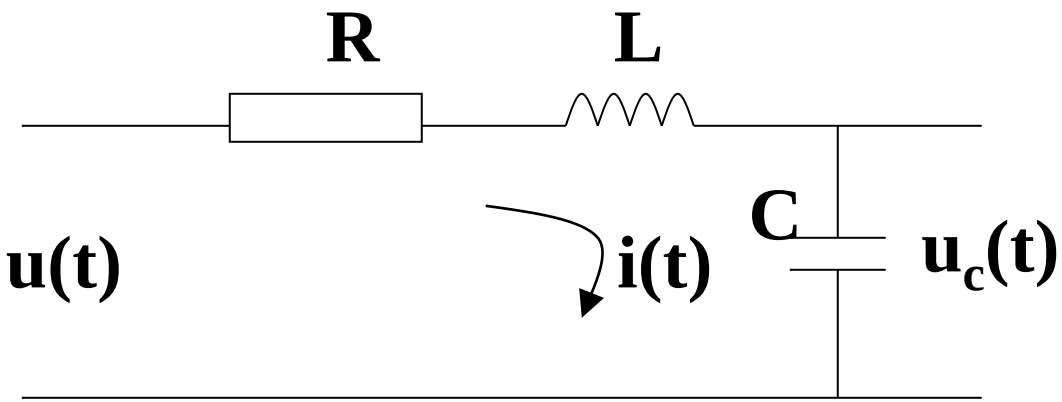
How to establish ODE of a control system

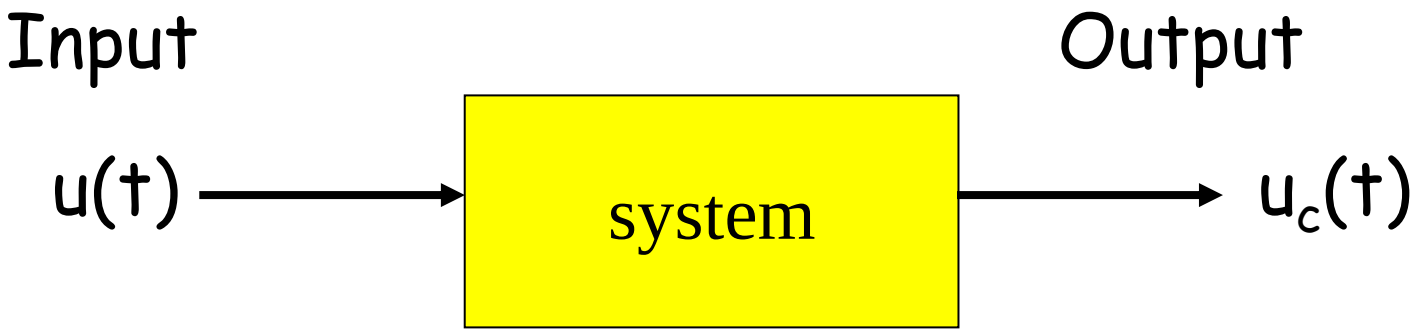
--- list differential equations according to the physical rules of each component;

--- obtain the differential equation sets by eliminating intermediate variables;

--- get the overall input-output differential equation of control system.

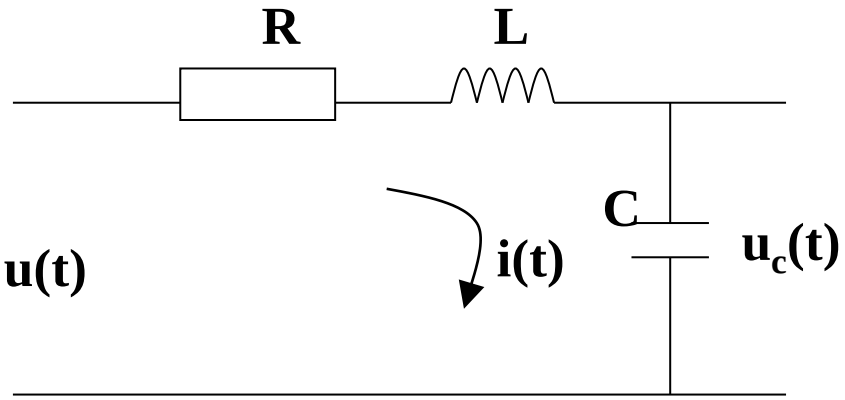
Example-1 RLC circuit





Defining the input and output according to which cause-effect relationship you are interested in.

According to Kirchhoff's Voltage Law



$$u(t) = Ri(t) + L\frac{di(t)}{dt} + u_c(t) - - - (1)$$

$$u_C(t) = \frac{1}{C}\int i(t)dt - - - (2)$$

$$\Rightarrow \quad i(t) = C\frac{du_C(t)}{dt}$$

$$u(t) = RC\frac{du_C(t)}{dt} + LC\frac{d^2u_C(t)}{dt^2} + u_C(t)$$

- It is re-written in standard form as

$$LC\ddot{u}_C(t) + RC\dot{u}_C(t) + u_C(t) = u(t)$$



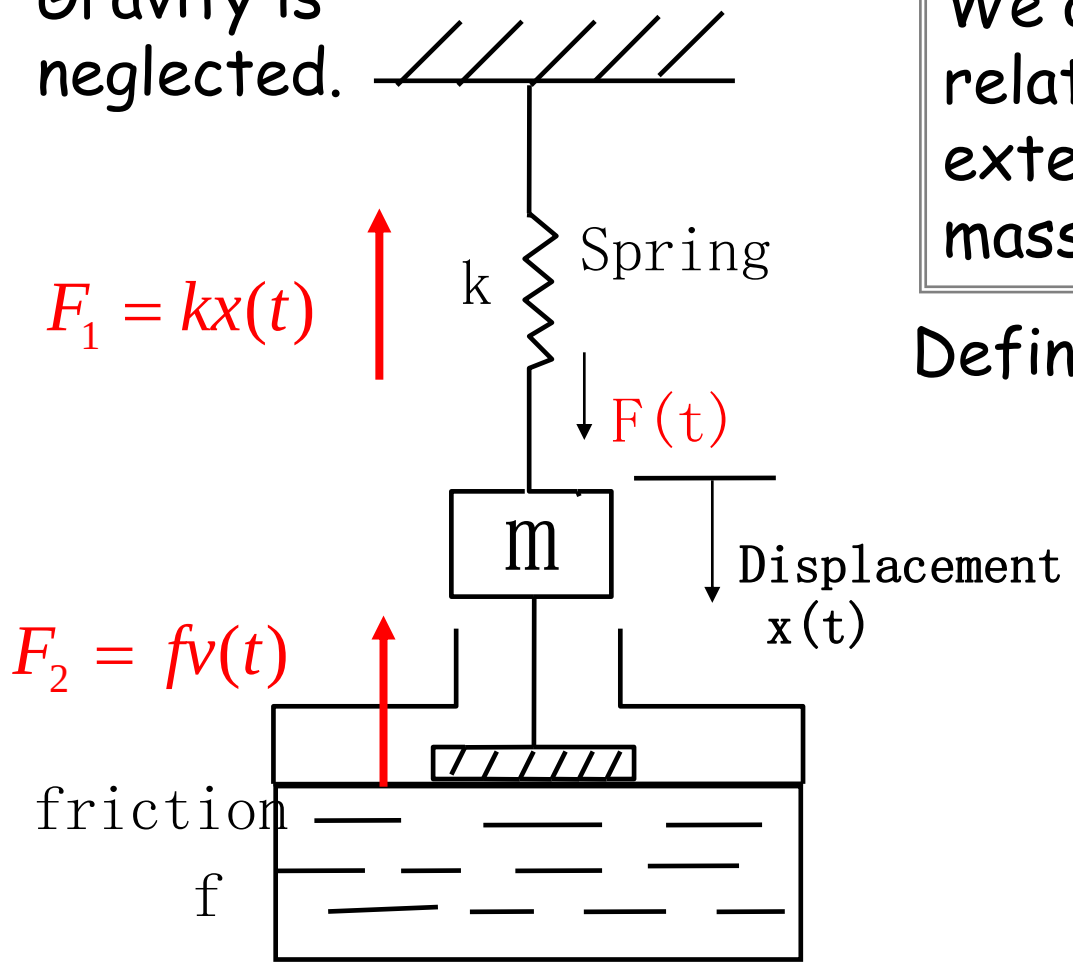
$$\ddot{u}_C(t) + \frac{R}{L}\dot{u}_C(t) + \frac{1}{LC}u_C(t) = \frac{1}{LC}u(t)$$

Generally, we set

- the output on the left side of the equation
- the input on the right side
- the input is arranged from the highest order to the lowest order

Example-2 Mass-spring-friction system

Gravity is neglected.



We are interested in the relationship between external force F(t) and mass displacement x(t)

Define: input—F(t); output---x(t)

$$\sum F = ma$$

$$ma = F - F_1 - F_2$$

$$v = \frac{dx(t)}{dt}, \quad a = \frac{d^2x(t)}{dt^2}$$

By eliminating intermediate variables, we obtain the overall input-output differential equation of the mass-spring-friction system.

$$m\ddot{x}(t) + f\dot{x}(t) + kx(t) = F(t)$$

Recall the RLC circuit system

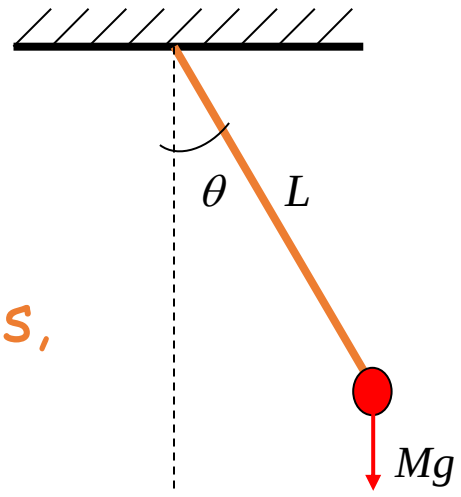
$$LC\ddot{u}_C(t) + RC\dot{u}_C(t) + u_C(t) = u(t)$$

These formulas are similar, that is, we can use the same mathematical model to describe a class of systems that are physically absolutely different but share the same Motion Law.

Example-3 nonlinear system

- In reality, most systems are indeed nonlinear, e.g. the pendulum system, which is described by nonlinear differential equations.

$$ML\frac{d^2\theta}{dt^2} + Mg\sin\theta(t) = 0$$

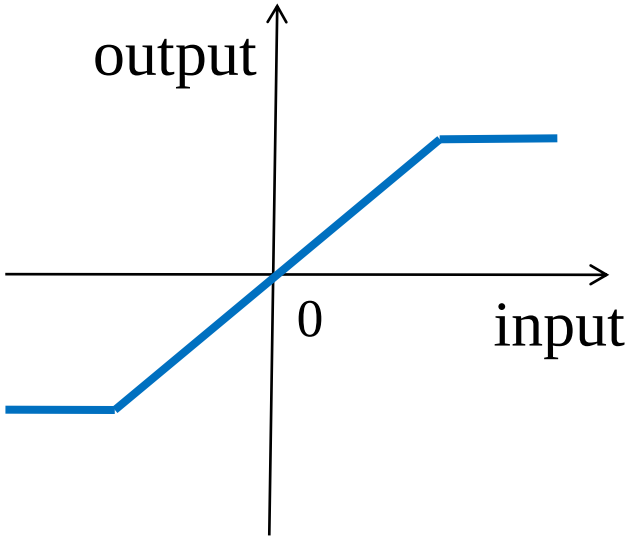


- It is difficult to analyze nonlinear systems, however, we can linearize the nonlinear system near its equilibrium point under certain conditions.

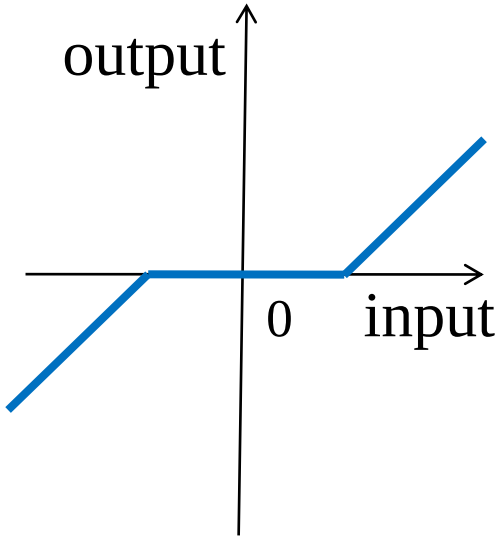
$$ML\frac{d^2\theta}{dt^2} + Mg\theta(t) = 0 \quad (\text{when } \theta \text{ is small})$$

Linearization of nonlinear differential equations

Several typical nonlinear characteristics in control system.



Saturation (Amplifier)



Dead-zone (Motor)

Methods of linearization

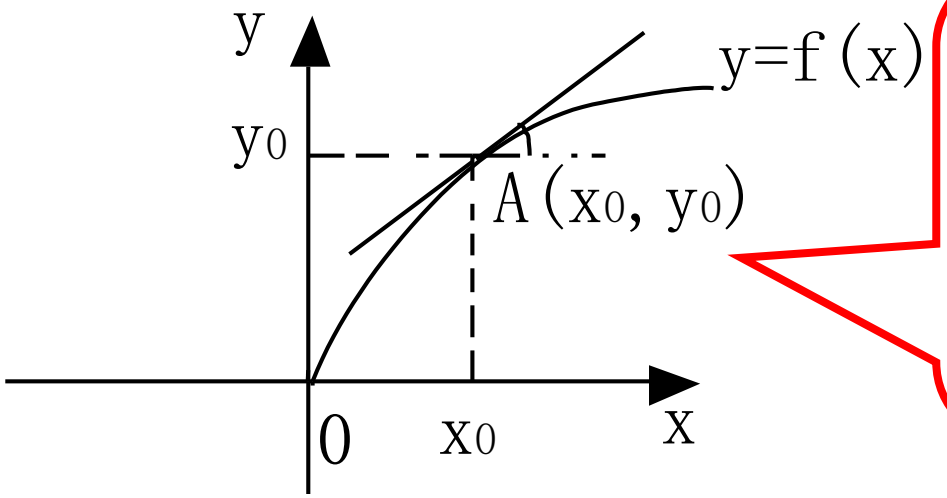
(1) Weak nonlinearity neglected

If the nonlinearity of the component is not within its linear working region, its effect on the system is weak and can be neglected.

(2) Small perturbation/error method

Assumption: In the system control process, there are just small changes around the equilibrium point in the input and output of each component.

This assumption is reasonable in many practical control systems: in closed-loop control system, once the deviation occurs, the control mechanism will reduce or eliminate it. Consequently, all the components can work around the equilibrium point.



A(x0,y0) is equilibrium point. Expanding the nonlinear function y=f(x) into a Taylor series about A(x0,y0) yields

$$y = f(x) = y_0 + \left.\frac{dy}{dx}\right|_{x_0} (x - x_0) + \left.\frac{1}{2!}\frac{d^2y}{dx^2}\right|_{x_0} (x - x_0)^2 + \cdots\cdots$$

The input and output only have small variance around the equilibrium point.

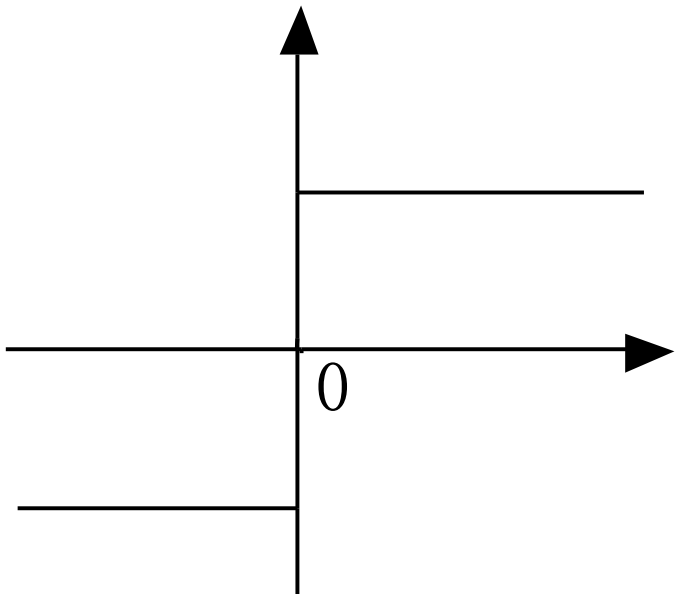
$$\Delta x = (x - x_0), (\Delta x)^n \to 0$$

$$y = y_0 + \left.\frac{dy}{dx}\right|_{x_0} (x - x_0)$$

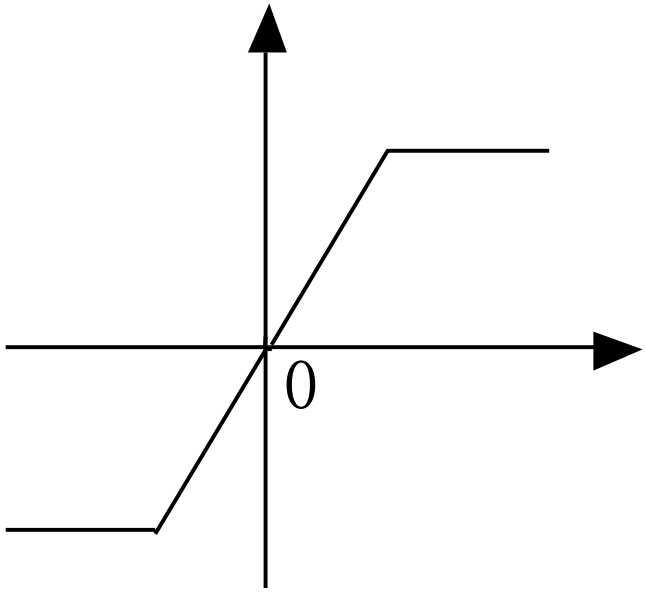
$$\Delta y = k\Delta x$$

This is linearized model of the nonlinear component.

Note: this method is only suitable for systems with weak nonlinearity.



Relay

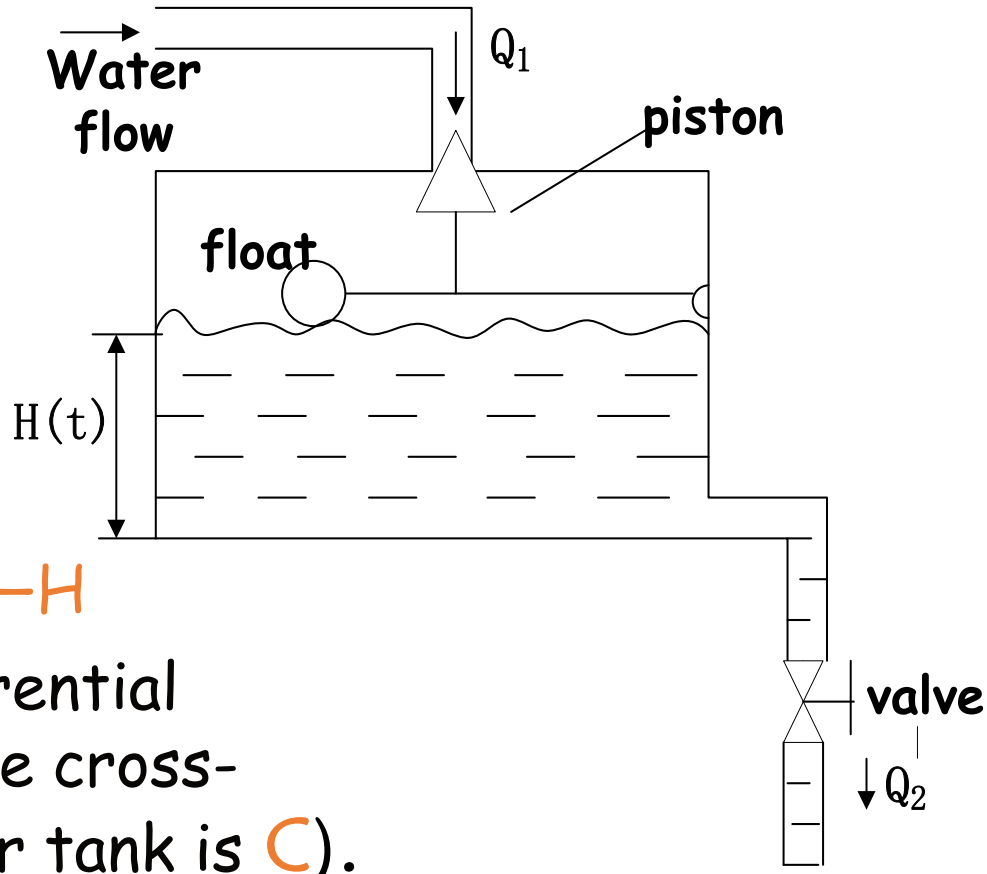


Saturation

For systems with strong nonlinearity, we cannot use such linearization method.

Example-4 modeling a nonlinear system

- Flash toilet



Q_1: inflow per unit time

Q_2: outflow per unit time

$Q_{10}=Q_{20}=0$

Initial water level: H_0

Define: Input—Q_1,Output—H

Problem: Derive the differential equation of water tank (the cross-sectional area of the water tank is C).

Solution: Outflow or inflow within *dt* time should be equal to the total amount of water change *(Q1-Q2) dt* , that is:

$$CdH = (Q_1 - Q_2)dt$$

According to 'Torricelli Theorem', the water yield is in direct proportion to the square root of the height of water level, thus:

$$Q_2 = \frac{\sqrt{H}}{R'}$$

R ' is a scale coefficent.

It is obvious that this formula is nonlinear. On the basis of the Taylor Series expansion of functions around operation points (Q_{10},H_0), we have

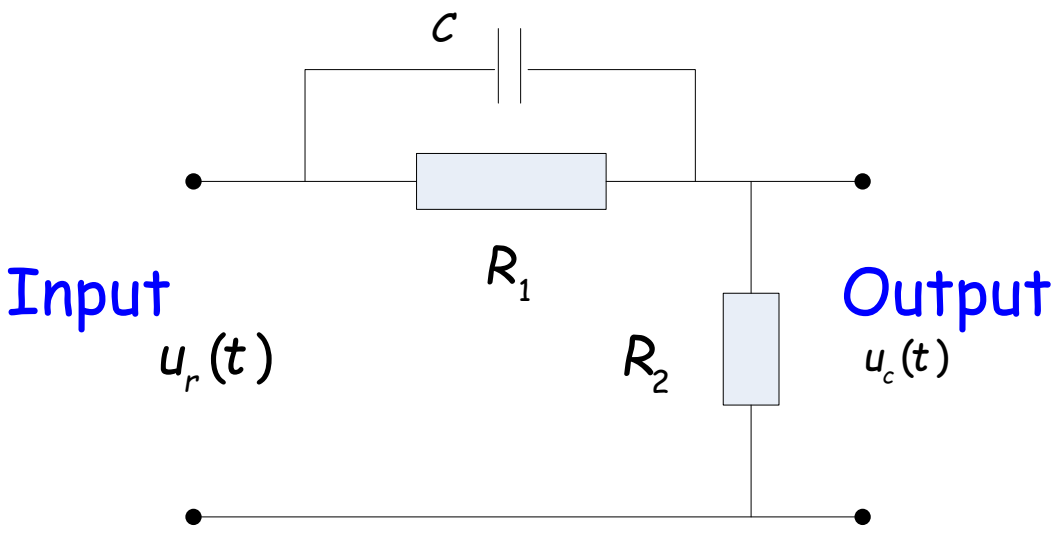
$$Q_2 = \frac{1}{2\sqrt{H_0}R'} \cdot H = \frac{H}{R},$$

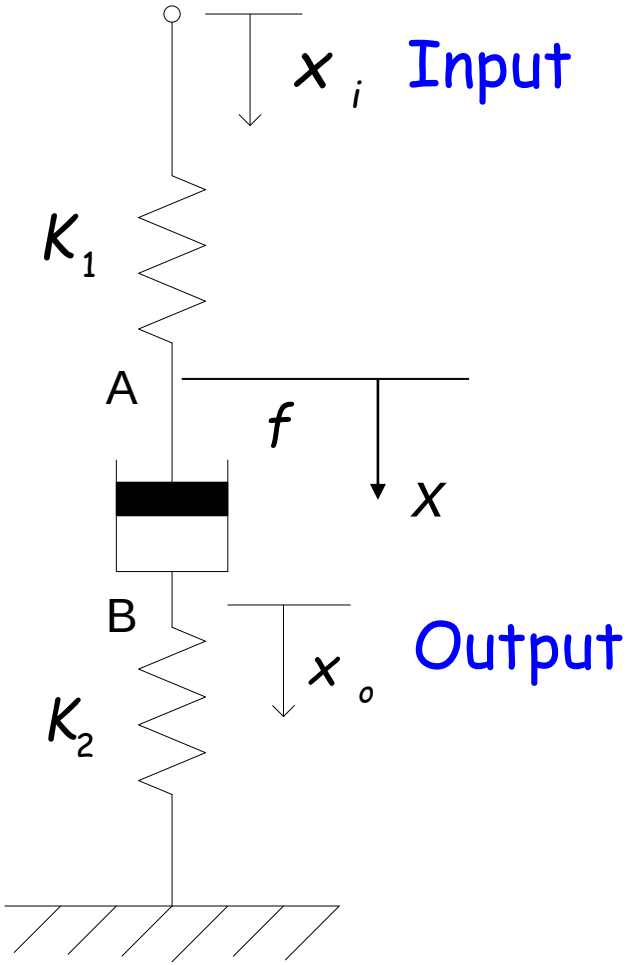
- Therefore, the Linearized Differential Equations of the water tank is:

$$RC\frac{dH}{dt} + H = RQ_1$$

Exercise

- E1. Please build the differential equations of the following two systems.





Solutions.

(1) RC circuit

$$\begin{cases} R_1 i_1 = \dfrac{1}{C}\int i_2 dt \\ i = i_1 + i_2 \\ u_c = R_2 i \\ u_r = R_1 i_1 + u_c \end{cases} \quad \Rightarrow \quad R_1 R_2 C \frac{du_c}{dt} + (R_1 + R_2) u_c = R_1 R_2 C \frac{du_r}{dt} + R_2 u_r$$

(2) Mass-spring system

$$\begin{cases} K_1(x_i - x) = f(\dot{x} - \dot{x}_o) \\ K_2 x_o = f(\dot{x} - \dot{x}_o) \end{cases} \quad \Rightarrow \quad f(K_1 + K_2)\frac{dx_o}{dt} + K_1 K_2 x_o = K_1 f \frac{dx_i}{dt}$$

Transfer function

Solving Differential Equations

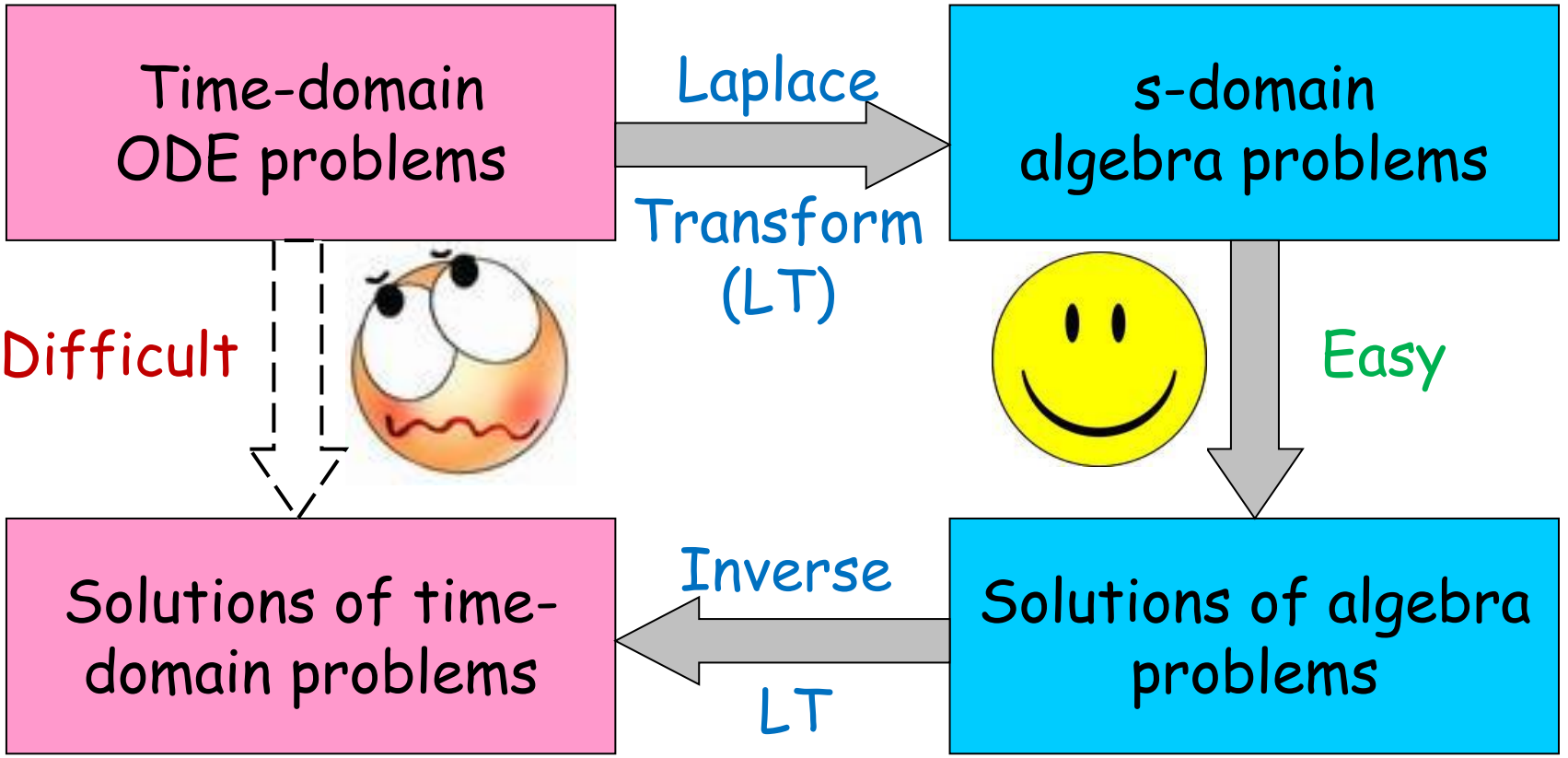
Example

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} = 2t + 1$$

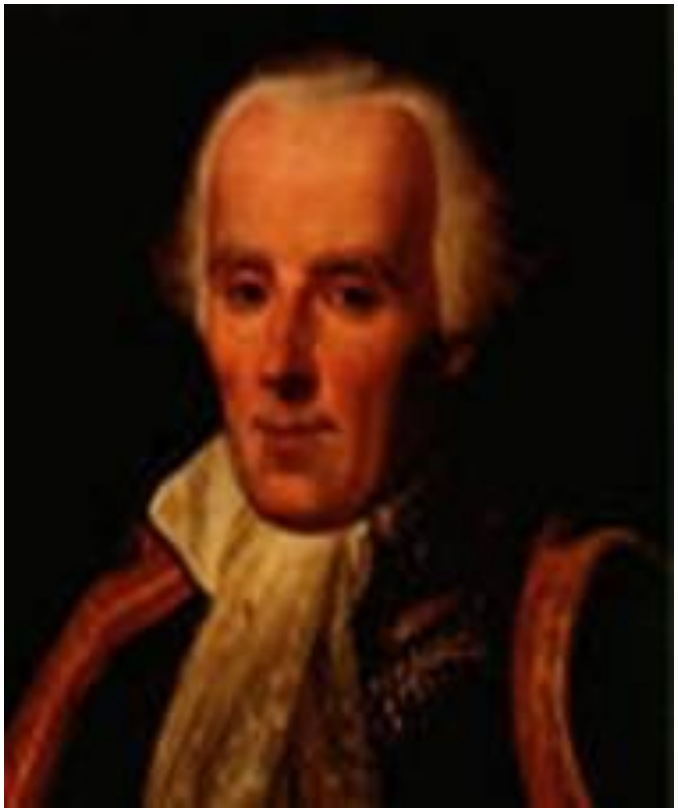
Solving linear differential equations with constant coefficients:

- Find the general solution (involving solving the characteristic equation)
- Find a particular solution of the complete equation (involving constructing the family of a function)

WHY need LAPLACE transform?



Laplace Transform



Laplace, Pierre-Simon
1749-1827

The Laplace transform of a function $f(t)$ is defined as

$$F(s) = \mathcal{L}\left[f(t) \right]$$

$$= \int_0^\infty f(t) e^{-st} dt$$

where $s = \sigma + j\omega$ is a complex variable.

Examples

- Step signal: $f(t)=A$

$$F(s)=\int_0^\infty f(t)e^{-st}dt=\int_0^\infty Ae^{-st}dt=-\frac{A}{s}e^{-st}\Big|_0^\infty=\frac{A}{s}$$

- Exponential signal $f(t)=e^{-at}$

$$F(s)=\int_0^\infty e^{-at}e^{-st}dt=-\frac{1}{s+a}e^{-(a+s)t}\Big|_0^\infty=\frac{1}{s+a}$$

Laplace transform table

f(t)	F(s)	f(t)	F(s)
δ (t)	1	$\sin wt$	$\frac{w}{s^2 + w^2}$
1(t)	$\frac{1}{s}$	$\cos wt$	$\frac{s}{s^2 + w^2}$
t	$\frac{1}{s^2}$	$e^{-at} \sin wt$	$\frac{w}{(s+a)^2 + w^2}$
e^{-at}	$\frac{1}{s+a}$	$e^{-at} \cos wt$	$\frac{s+a}{(s+a)^2 + w^2}$

Properties of Laplace Transform

(1) Linearity

$$\mathcal{L}[af_1(t)+bf_2(t)]=a\mathcal{L}[f_1(t)]+b\mathcal{L}[f_2(t)]$$

(2) Differentiation

$$\mathcal{L}\left[\frac{df(t)}{dt}\right]=sF(s)-f(0)$$

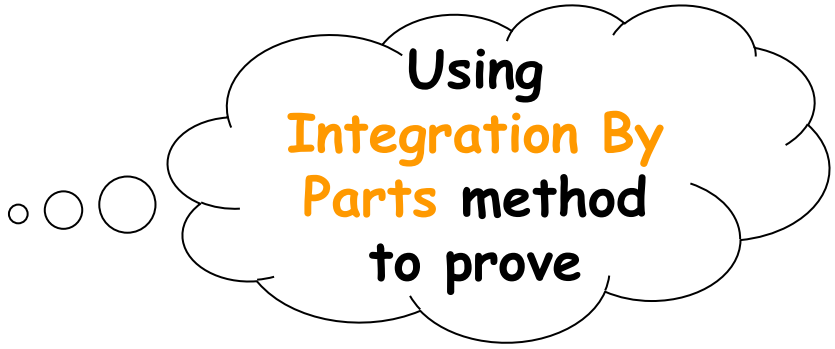
Using Integration By Parts method to prove

where f(0) is the initial value of f(t).

$$\mathcal{L}\left[\frac{d^nf(t)}{dt^n}\right]=s^nF(s)-s^{n-1}f(0)-s^{n-2}f^{(1)}(0)-\cdots-f^{(n-1)}(0)$$

(3) Integration

$$\mathcal{L}\left[\int_0^t f(\tau)d\tau\right] = \frac{F(s)}{s}$$



$$\mathcal{L}\left[\int_o^{t_1}\int_o^{t_2}\cdots\int_o^{t_n} f(\tau)d\tau dt_1 dt_2\cdots dt_{n-1}\right] = \frac{F(s)}{s^n}$$

(4) Final-value Theorem

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

The final-value theorem relates the steady-state behavior of f(t) to the behavior of sF(s) in the neighborhood of s=0

(5) Initial-value Theorem

$$\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$$

(6)Shifting Theorem:

a. shift in time (real domain)

$$\mathcal{L}[f(t-\tau)] = e^{-\tau s}F(s)$$

b. shift in complex domain

$$\mathcal{L}[e^{at}f(t)] = F(s-a)$$

(7) Real convolution (Complex multiplication) Theorem

$$\mathcal{L}[\int_0^t f_1(t-\tau)f_2(\tau)d\tau] = F_1(s)\cdot F_2(s)$$

Inverse Laplace transform

Definition: Inverse Laplace transform, denoted by $\mathcal{L}^{-1}[F(s)]$ is given by

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi \cdot j} \int_{C-j\infty}^{C+j\infty} F(s)e^{st} ds, (t > 0)$$

where C is a real constant。

Note: The inverse Laplace transform operation involving rational functions can be carried out using Laplace transform table and partial-fraction expansion.

Partial-Fraction Expansion method for finding Inverse Laplace Transform

$$F(s) = \frac{N(s)}{D(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \quad (m < n)$$

If *F(s)* is broken up into components

$$F(s) = F_1(s) + F_2(s) + \ldots + F_n(s)$$

If the inverse Laplace transforms of components are readily available, then

$$\mathcal{L}^{-1}\left[F(s)\right] = \mathcal{L}^{-1}\left[F_1(s)\right] + \mathcal{L}^{-1}\left[F_2(s)\right] + \ldots + \mathcal{L}^{-1}\left[F_n(s)\right]$$

$$= f_1(t) + f_2(t) + \ldots + f_n(t)$$

Poles and zeros

- Poles
  - A complex number s_0 is said to be a pole of a complex variable function $F(s)$ if $F(s_0)=\infty$.
- Zeros
  - A complex number s_0 is said to be a zero of a complex variable function $F(s)$ if $F(s_0)=0$.

Examples:

$$\frac{(s-1)(s+2)}{(s+3)(s+4)}$$ poles: -3, -4; zeros: 1, -2

$$\frac{s+1}{s^2+2s+2}$$ poles: -1+j, -1-j; zeros: -1

Case 1: F(s) has simple real poles

$$F(s)=\frac{N(s)}{D(s)}=\frac{b_0 s^m+b_1 s^{m-1}+\cdots+b_{m-1}s+b_m}{s^n+a_1 s^{n-1}+\cdots+a_{n-1}s+a_n}$$

Partial-Fraction Expansion

$$=\frac{c_1}{s+p_1}+\frac{c_2}{s+p_2}+\cdots+\frac{c_n}{s+p_n}$$

where $s=-p_i\,(i=1,2,\cdots,n)$ are roots of $D(s)=0$, and

$$c_i=\left[\frac{N(s)}{D(s)}(s+p_i)\right]\Big|_{s=-p_i}$$

Inverse LT

$$f(t)=c_1e^{-p_1t}+c_2e^{-p_2t}+\ldots+c_ne^{-p_nt}$$

Parameters p_k give shape and numbers c_k give magnitudes.

Example 1

Partial-Fraction Expansion

$$F(s)=\frac{1}{(s+1)(s-2)(s+3)}=\frac{c_1}{s+1}+\frac{c_2}{s-2}+\frac{c_3}{s+3}$$

$$c_1=\left[\frac{1}{(s+1)(s-2)(s+3)}\cdot(s+1)\right]\Bigg|_{s=-1}=-\frac{1}{6}$$

$$c_2=\left[\frac{1}{(s+1)(s-2)(s+3)}\cdot(s-2)\right]\Bigg|_{s=2}=\frac{1}{15}$$

$$c_3=\left[\frac{1}{(s+1)(s-2)(s+3)}\cdot(s+3)\right]\Bigg|_{s=-3}=\frac{1}{10}$$

$$\therefore F(s)=-\frac{1}{6}\cdot\frac{1}{s+1}+\frac{1}{15}\cdot\frac{1}{s-2}+\frac{1}{10}\cdot\frac{1}{s+3}$$

$$\therefore f(t)=-\frac{1}{6}e^{-t}+\frac{1}{15}e^{2t}+\frac{1}{10}e^{-3t}$$

Case 2: F(s) has simple complex-conjugate poles

Example 2

$$\ddot{y}(t)+4\dot{y}(t)+5y(t)=0, y(0)=\dot{y}(0)=1$$

Laplace transform

$$s^2Y(s)-sy(0)-\dot{y}(0)+4sY(s)-4y(0)+5Y(s)=0$$

Applying initial conditions

$$\therefore Y(s)=\frac{s+5}{s^2+4s+5}=\frac{s+5}{(s+2)^2+1}=\frac{s+2+3}{(s+2)^2+1}$$

Partial-Fraction Expansion

$$=\frac{s+2}{(s+2)^2+1}+\frac{3}{(s+2)^2+1}$$

Inverse Laplace transform

$$y(t)=e^{-2t}\cos t+3e^{-2t}\sin t$$

Case 3: F(s) has multiple-order poles

$$F(s)=\frac{N(s)}{D(s)}=\frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_{n-l})(s+p_i)^l}$$

$$=\frac{c_1}{s+p_1}+\cdots+\frac{c_{n-l}}{s+p_{n-l}}+\frac{b_l}{(s+p_i)^l}+\frac{b_{l-1}}{(s+p_i)^{l-1}}+\cdots+\frac{b_1}{s+p_i}$$

Simple poles — Multi-order poles

The coefficients $c_1,\cdots,c_{n-l}$ of simple poles can be calculated as Case 1;

The coefficients corresponding to the multi-order poles are determined as

$$b_l=\left[F(s)\cdot(s+p_i)^l\right]\Big|_{s=-p_i},\ b_{l-1}=\left\{\frac{d}{ds}\left[F(s)\cdot(s+p_i)^l\right]\right\}\Big|_{s=-p_i},\cdots,$$

$$b_{l-m}=\frac{1}{m!}\left\{\frac{d^m}{ds}\left[\frac{N(s)}{D(s)}(s+p_i)^l\right]\right\}\Big|_{s=-p_i},\ b_1=\frac{1}{(l-1)!}\left\{\frac{d^{l-1}}{ds}\left[\frac{N(s)}{D(s)}(s+p_i)^l\right]\right\}\Big|_{s=-p_i}$$

Example 3

Solve the following differential equation

$$y^{(3)} + 3\ddot{y} + 3\dot{y} + y = 1, y(0) = \dot{y}(0) = \ddot{y}(0) = 0$$

Laplace transform:

$$s^3Y(s) - s^2y(0) - s\dot{y}(0) - \ddot{y}(0) + 3s^2Y(s) - 3sy(0) - 3\dot{y}(0)$$

$$+ 3sY(s) - 3y(0) + Y(s) = \frac{1}{s}$$

Applying initial conditions:

$$Y(s) = \frac{1}{s(s+1)^3}$$

$s = -1$ is a 3-order pole

Partial-Fraction Expansion

$$Y(s) = \frac{c_1}{s} + \frac{b_3}{(s+1)^3} + \frac{b_2}{(s+1)^2} + \frac{b_1}{s+1}$$

Determining coefficients:

$$c_1 = \frac{1}{s(s+1)^3} s \Big|_{s=0} = 1$$

$$b_3 = [\frac{1}{s(s+1)^3}(s+1)^3]_{s=-1} = -1 \qquad b_1 = \frac{1}{2!}(2s^{-3})\Big|_{s=-1} = -1$$

$$b_2 = \left\{\frac{d}{ds}[\frac{1}{s(s+1)^3}(s+1)^3]\right\}_{s=-1} = [\frac{d}{ds}(\frac{1}{s})]_{s=-1} = (-s^{-2})\Big|_{s=-1} = -1$$

$$\therefore Y(s) = \frac{1}{s} - \frac{1}{(s+1)^3} - \frac{1}{(s+1)^2} - \frac{1}{s+1}$$

Inverse Laplace transform:

$$y(t) = 1 - \frac{1}{2}t^2 e^{-t} - te^{-t} - e^{-t}$$

With aid of MATLAB

1. Laplace Transform

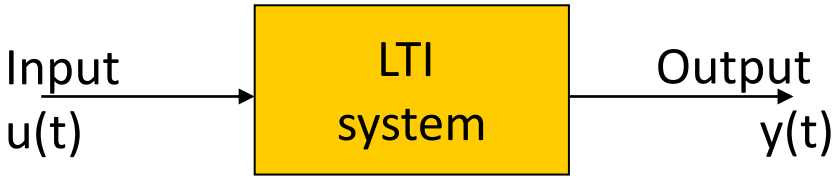
L=laplace(f)

2. Inverse Laplace Transform

F=ilaplace(L)

```
>> syms t
>> L=laplace(t)
L=
1/s^2
>> L=laplace(sin(t))
L=
1/(s^2+1)
>> F=ilaplace(L)
F=
sin(t)
```

Transfer function



Consider a linear system described by differential equation

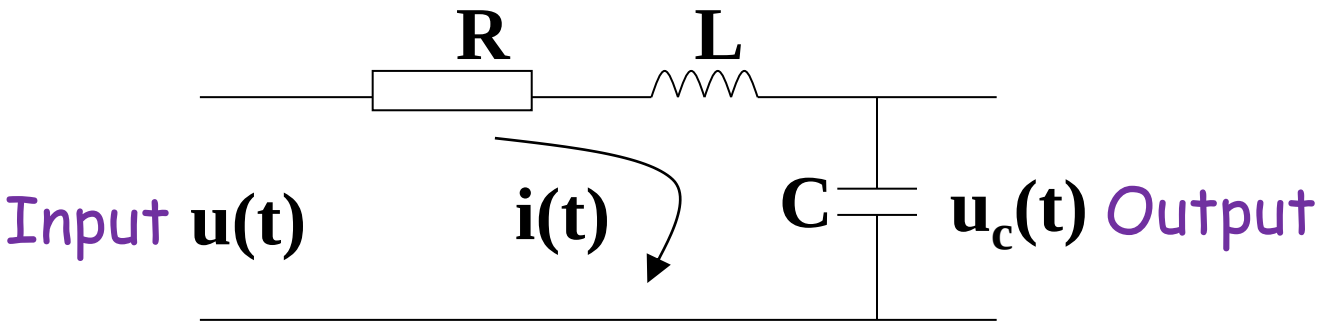
$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_0y(t)=\ b_mu^{(m)}(t)+b_{m-1}u^{(m-1)}(t)+\cdots+bu^{(1)}(t)+b_0u(t)$$

Assuming all initial conditions are zero, we get the transfer function (TF) of the system as

$$TF=G(s)=\left.\frac{\mathcal{L}[output\ y(t)]}{\mathcal{L}[input\ u(t)]}\right|_{zero\ initial\ condition}$$

$$=\frac{Y(s)}{U(s)}=\frac{b_ms^m+b_{m-1}s^{m-1}+\ldots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\ldots+a_1s+a_0}$$

Example 1. Find the transfer function of the RLC



Solution:

1) Writing the differential equation of the system according to physical law:

$$LC\ddot{u}_C(t) + RC\dot{u}_C(t) + u_C(t) = u(t)$$

2) Assuming all initial conditions are zero and applying Laplace transform

$$LCs^2U_c(s) + RCsU_c(s) + U_c(s) = U(s)$$

3) Calculating the transfer function $G(s)$ as

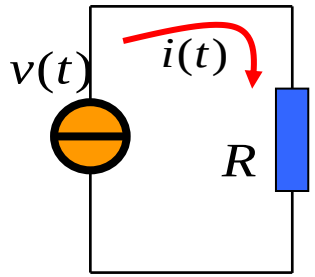
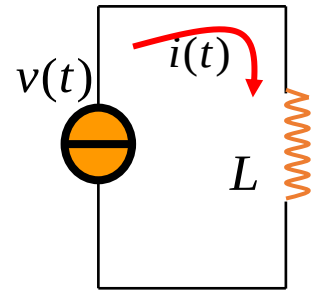
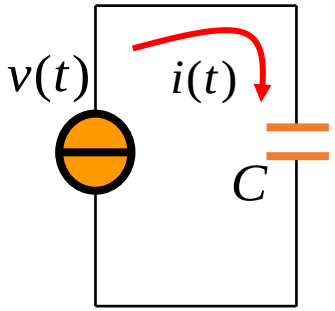
$$G(s) = \frac{U_c(s)}{U(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Exercise

- Find the transfer function of the following system :

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 4y(t) = u(t)$$

Transfer function of typical components

Component	ODE	TF
	$v(t) = Ri(t)$	$G(s) = \frac{V(s)}{I(s)} = R$
	$v(t) = L\frac{di(t)}{dt}$	$G(s) = \frac{V(s)}{I(s)} = sL$
	$v(t) = \frac{1}{C}\int_0^t i(\tau)d\tau$	$G(s) = \frac{V(s)}{I(s)} = \frac{1}{sC}$

Properties of transfer function

- The transfer function is defined only for linear time-invariant (LTI) systems, not for time-varying or nonlinear systems.
- All initial conditions of the system are set to zero.
- The transfer function is independent of the input of the system.
- The transfer function $G(s)$ is the Laplace transform of the unit impulse response $g(t)$.

How poles and zeros relate to system response

- Why we strive to obtain TF models?
- Why control engineers prefer to use TF model?
- How to use TF model to analyze and design control systems?

- we start from the relationship between the locations of zeros and poles of TF and the output responses of a system

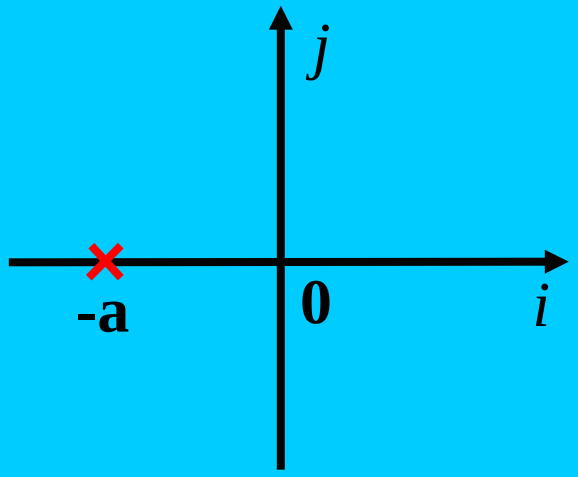
Transfer function

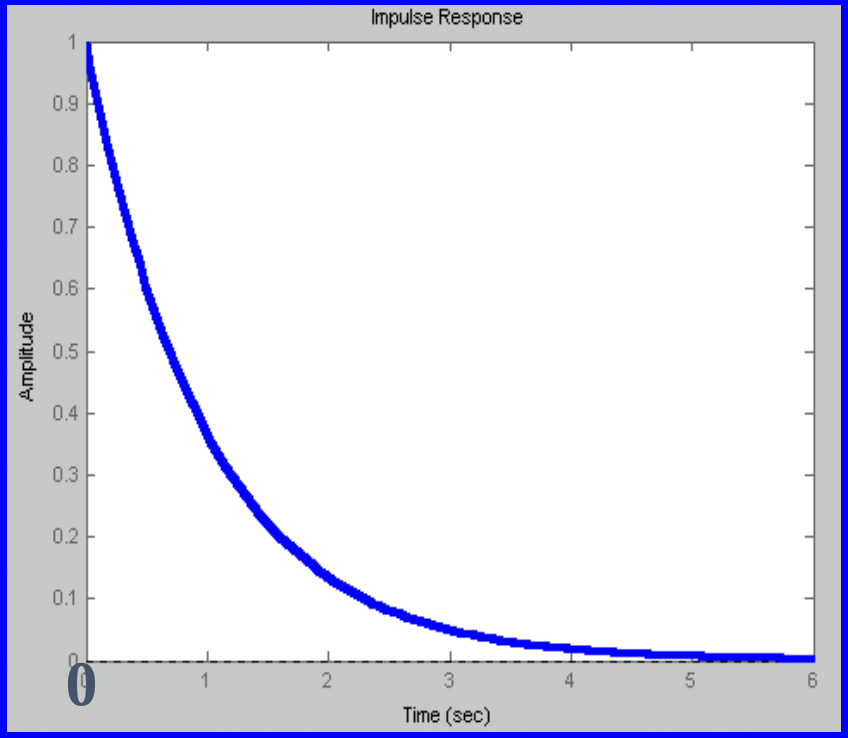
$$X(s) = \frac{A}{s + a}$$

Time-domain impulse response

$$x(t) = Ae^{-at}$$

Position of poles and zeros





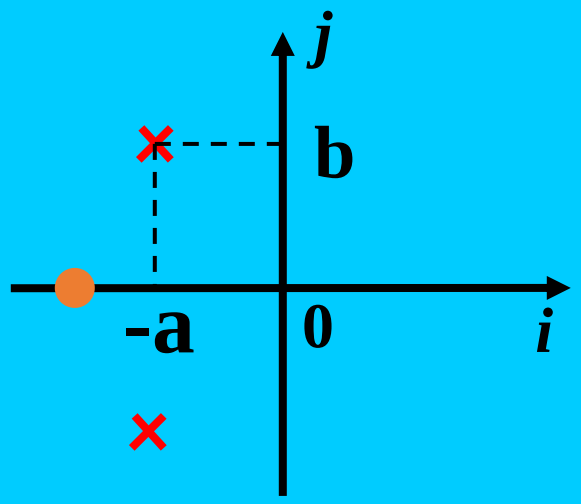
Transfer function

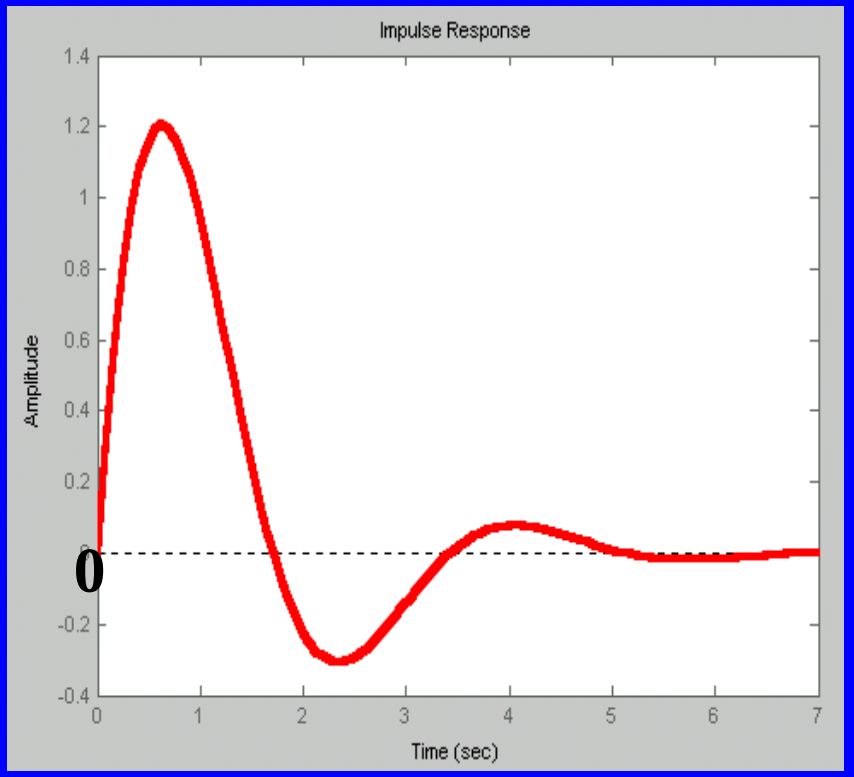
$$X(s) = \frac{A_1 s + B_1}{(s+a)^2 + b^2}$$

Time-domain impulse response

$$x(t) = Ae^{-at}\sin(bt + \phi)$$

Position of poles and zeros





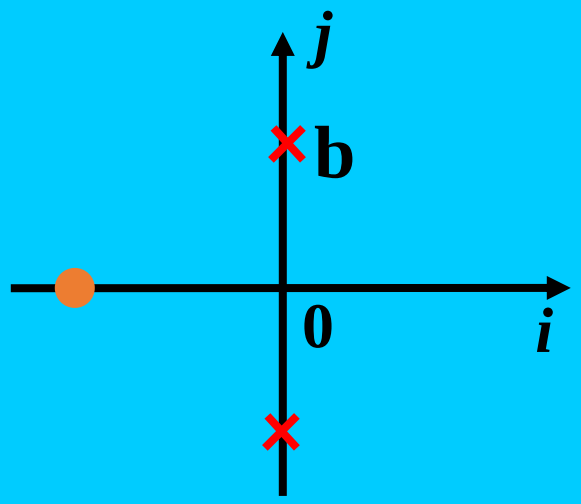
Transfer function

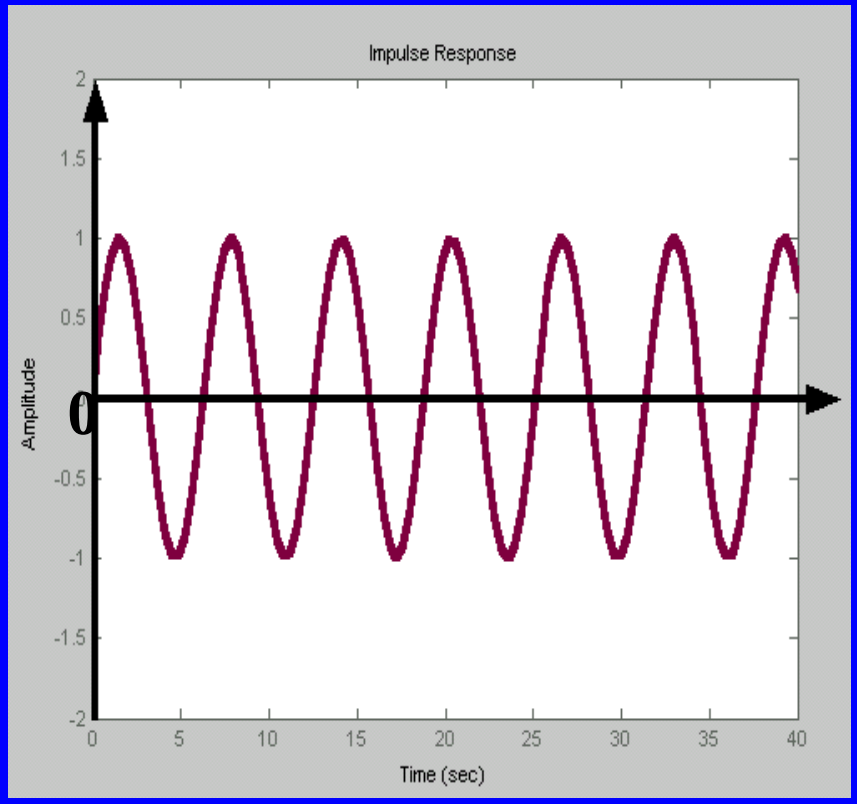
$$X(s) = \frac{A_1 s + B_1}{s^2 + b^2}$$

Time-domain impulse response

$$x(t) = A\sin(bt + \phi)$$

Position of poles and zeros





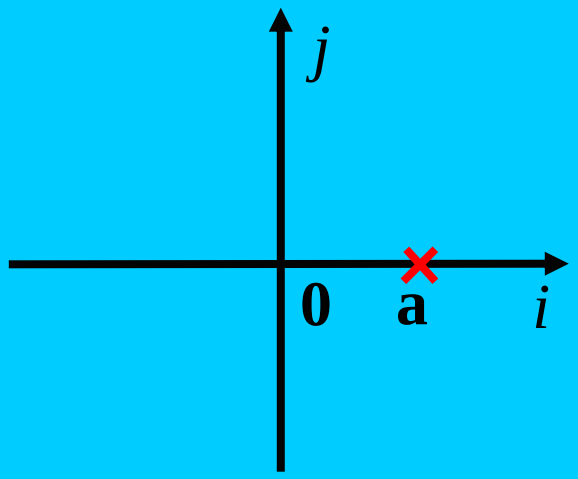
Transfer function

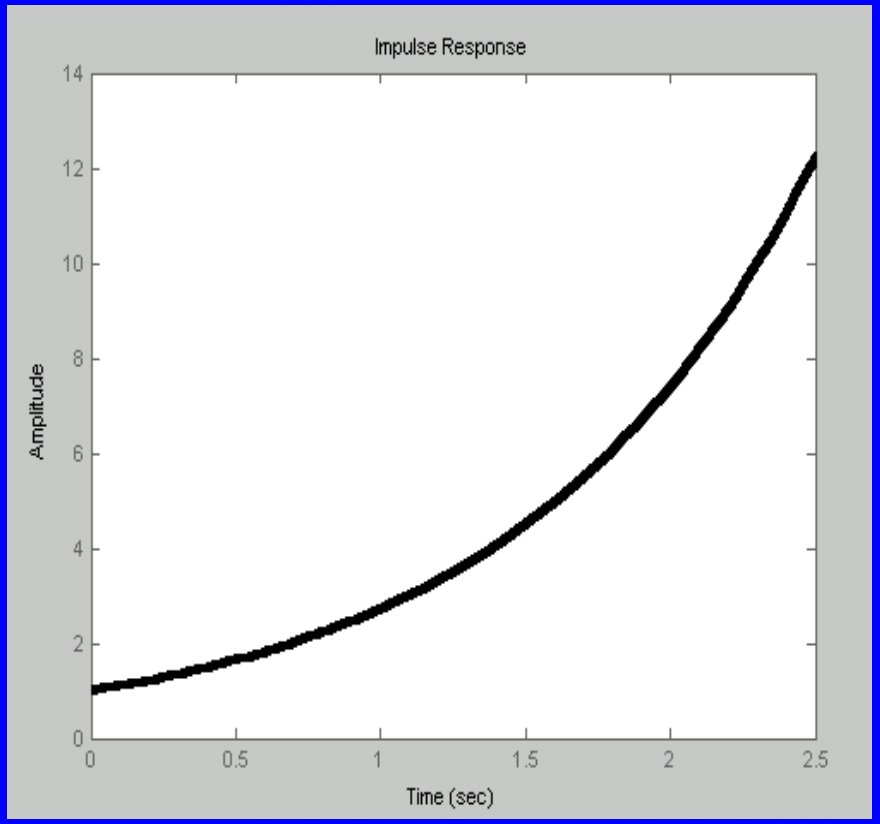
$$X(s) = \frac{A}{s - a}$$

Time-domain impulse response

$$x(t) = Ae^{at}$$

Position of poles and zeros





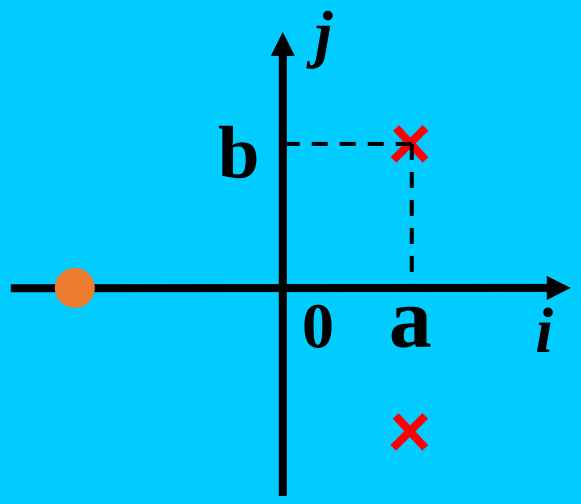
Transfer function:

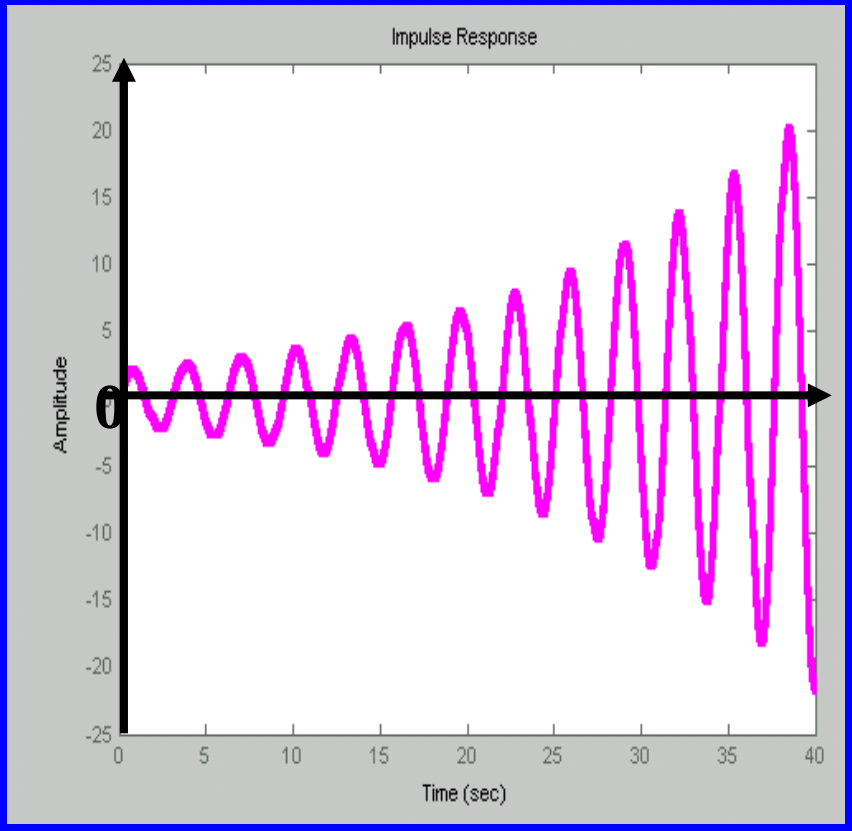
$$X(s) = \frac{A_1 s + B_1}{(s-a)^2 + b^2}$$

Time-domain dynamic response

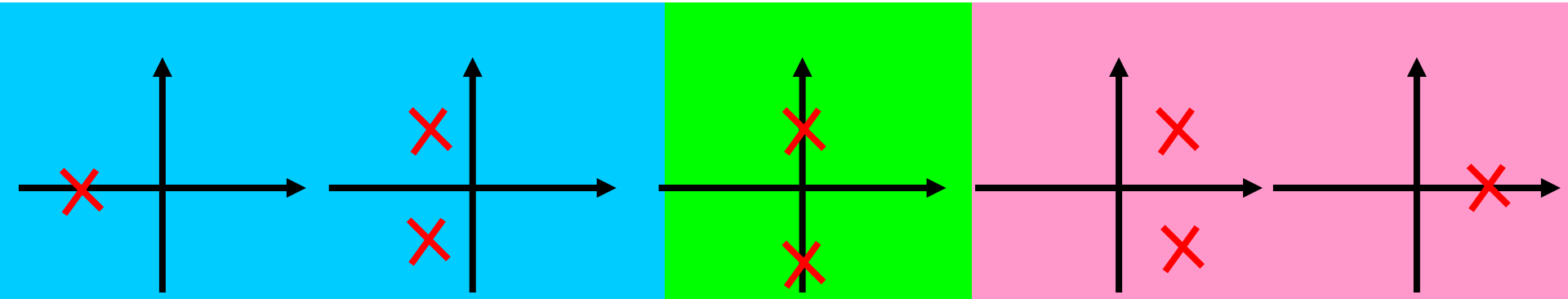
$$x(t) = Ae^{at}\sin(bt + \phi)$$

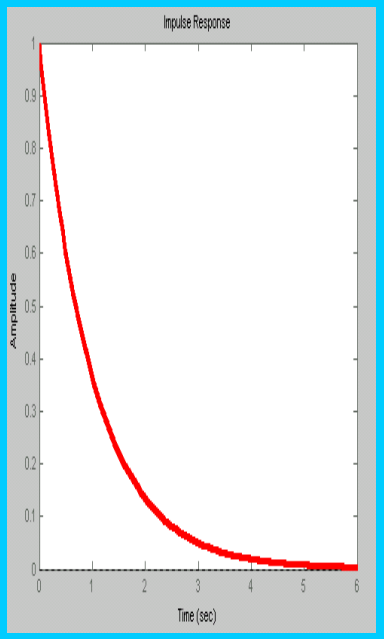
Position of poles and zeros

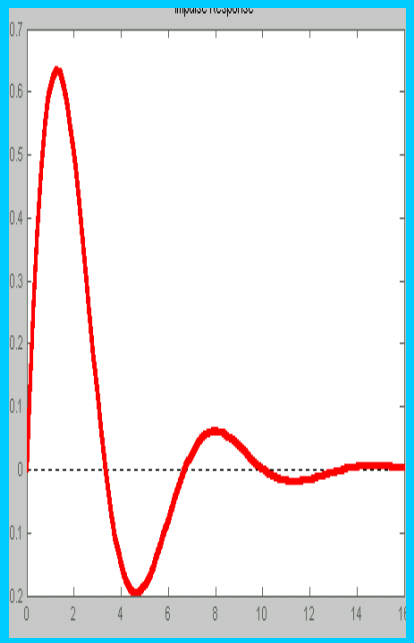


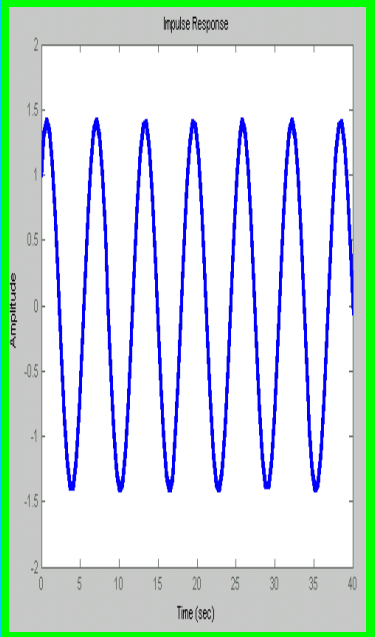


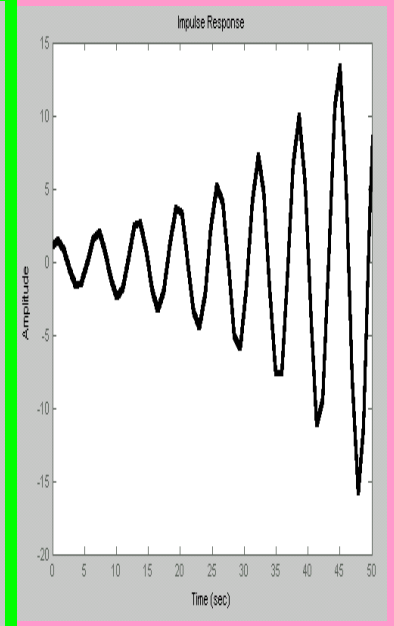
Summary of pole position & system dynamics

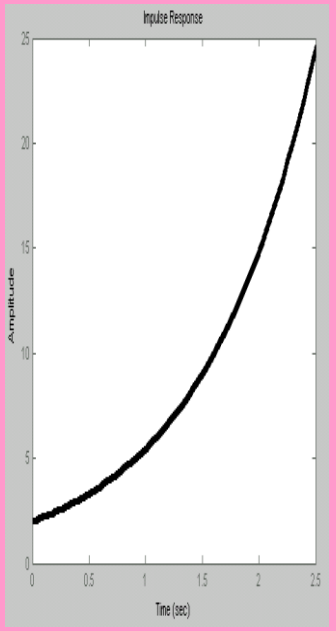












Characteristic equation

-obtained by setting the denominator polynomial of the transfer function equal to zero

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

Note: stability of linear single-input, single-output systems is completely governed by the roots of the characteristics equation.

Transfer function (TF) models in MATLAB

Suppose a linear SISO system with input u(t), output y(t), the transfer function of the system is

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \ldots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}$$

$$num = [b_m, b_{m-1}, \ldots, b_0]$$

$$den = [1, a_{n-1}, \ldots, a_0]$$

Descending power of s

TF in polynomial form

```
>> sys = tf(num,  den)
>> [num, den] = tfdata(sys)
```

TF in zero-pole form

```
>> sys = zpk(z, p, k)
>> [z, p, k] = zpkdata (sys)
```

Transform from polynomial form into zero-pole form

```
>> [z, p, k] = tf2zp(num, den)
```

Transform from zero-pole form into polynomial form

```
>> [num, den] = zp2tf(z, p, k)
```

Review questions

- What is the definition of "transfer function"?
- When defining the transfer function, what happens to initial conditions of the system?
- Does a nonlinear system have a transfer function?
- How does a transfer function of a LTI system relate to its impulse response?
- Define the characteristic equation of a linear system in terms of the transfer function.

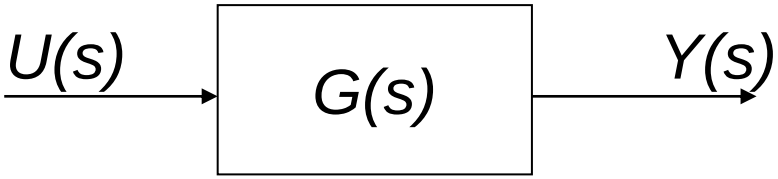
Block diagram and Signal-flow graph (SFG)

Block diagram

The transfer function relationship

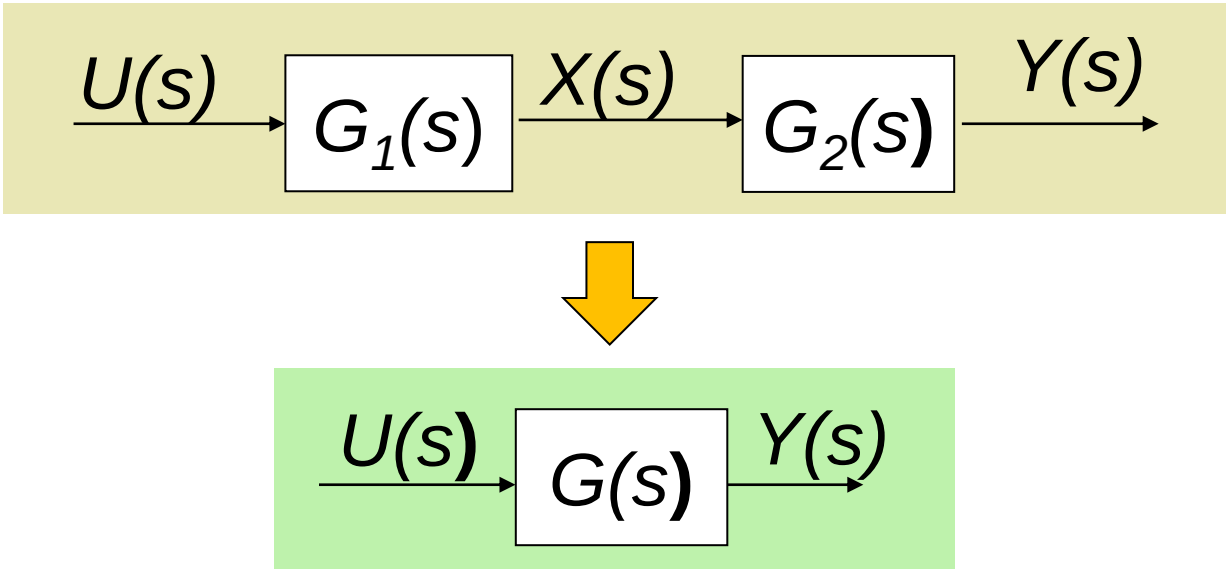
$$Y(s) = G(s)U(s)$$

can be graphically denoted through a block diagram.



Equivalent transform of block diagram

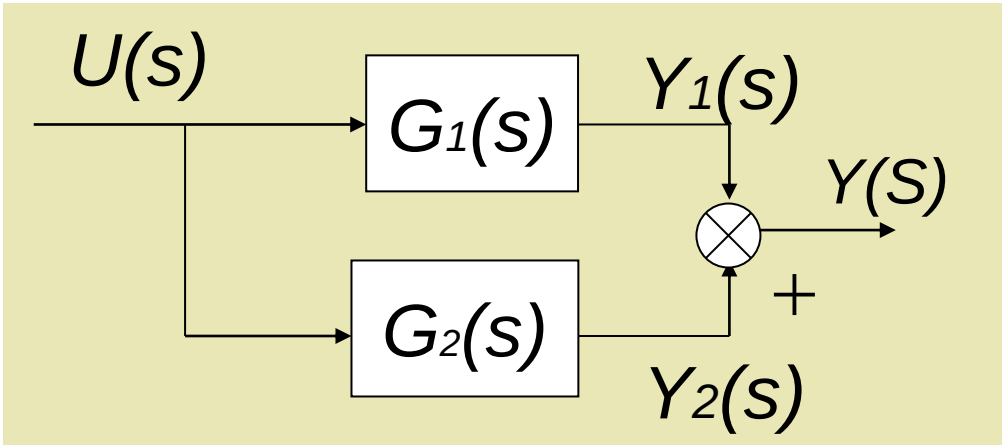
1 Connection in series

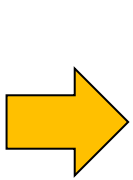


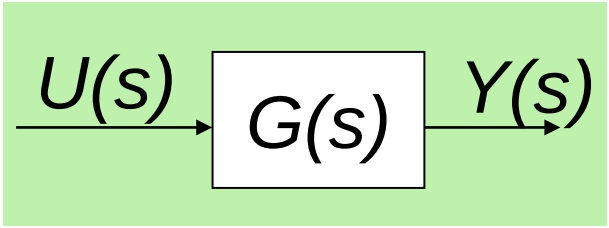
$G(s) = ?$

$$G(s) = \frac{Y(s)}{U(s)} = G_1(s) \cdot G_2(s)$$

2.Connection in parallel



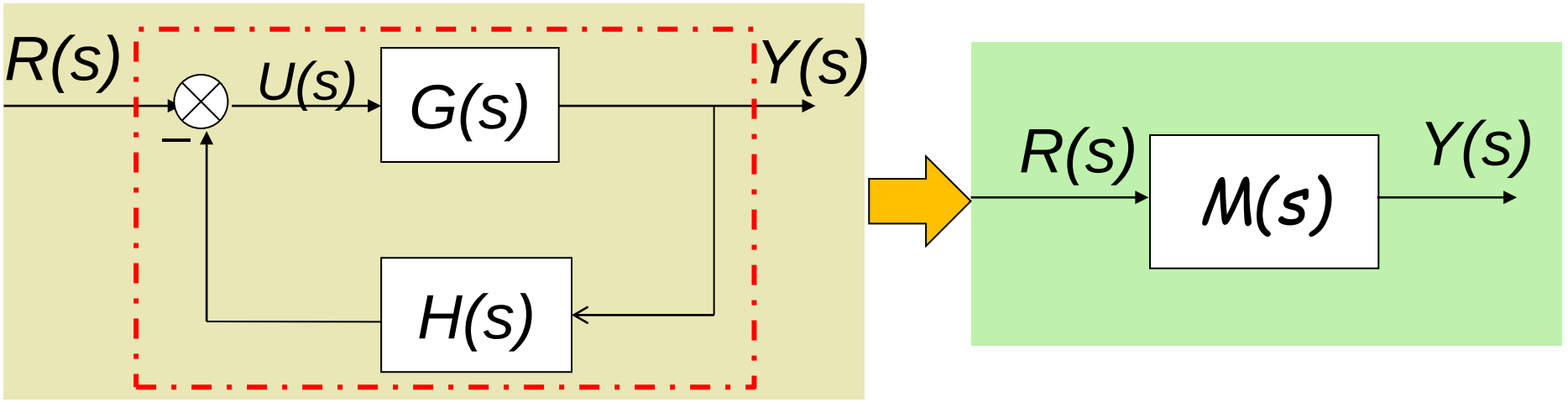




$G(s) = ?$

$$G(s) = \frac{Y(s)}{U(s)} = G_1(s) + G_2(s)$$

3. Negative feedback

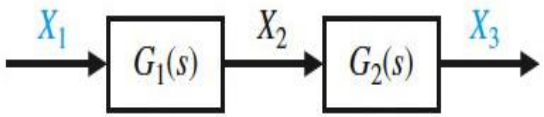
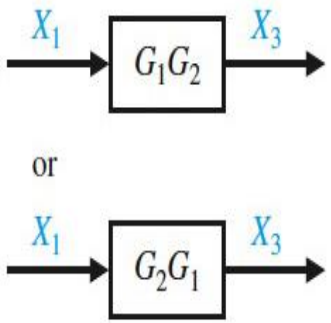
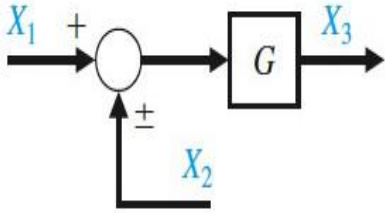
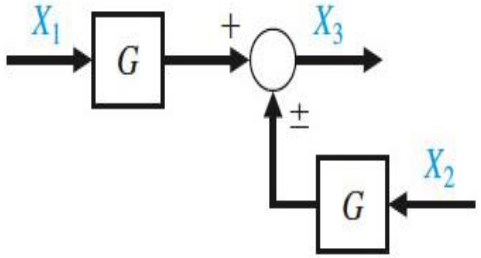
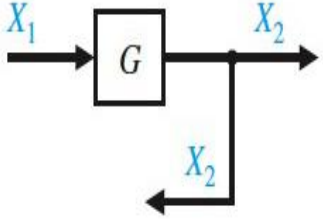
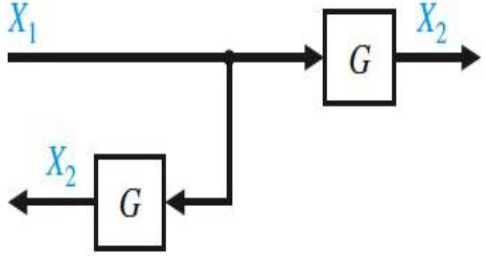


$$\begin{cases} Y(s) = G(s)U(s) \\ U(s) = R(s) - H(s)Y(s) \end{cases} \Rightarrow Y(s) = G(s)\left[R(s) - H(s)Y(s)\right]$$

Transfer function of a negative feedback system:

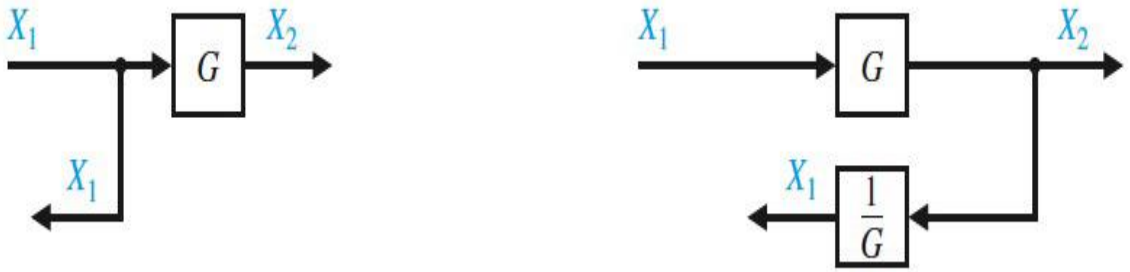
$$M(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{\text{gain of the forward path}}{1 + \text{gain of the loop}}$$

Block Diagram Reduction

Transformation	Original Diagram	Equivalent Diagram
1. Combining blocks in cascade		
2. Moving a summing point behind a block		
3. Moving a pickoff point ahead of a block		

Block Diagram Reduction

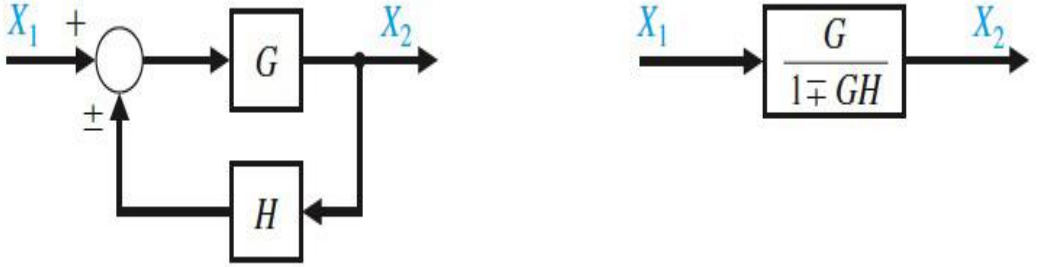
4. Moving a pickoff point behind a block



5. Moving a summing point ahead of a block



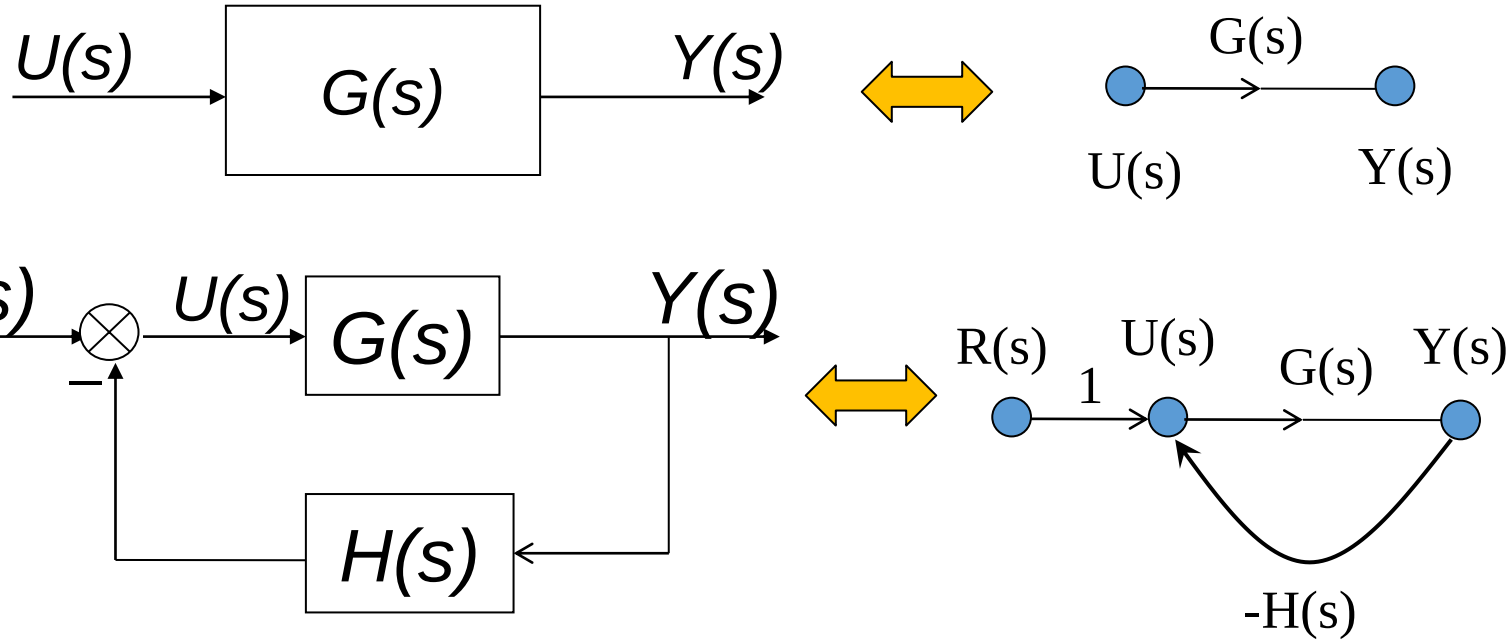
6. Eliminating a feedback loop



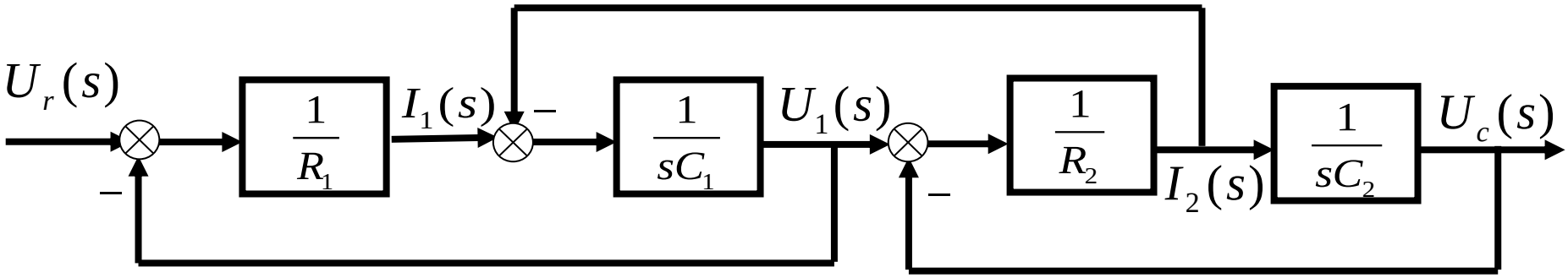
Signal flow graph (SFG)

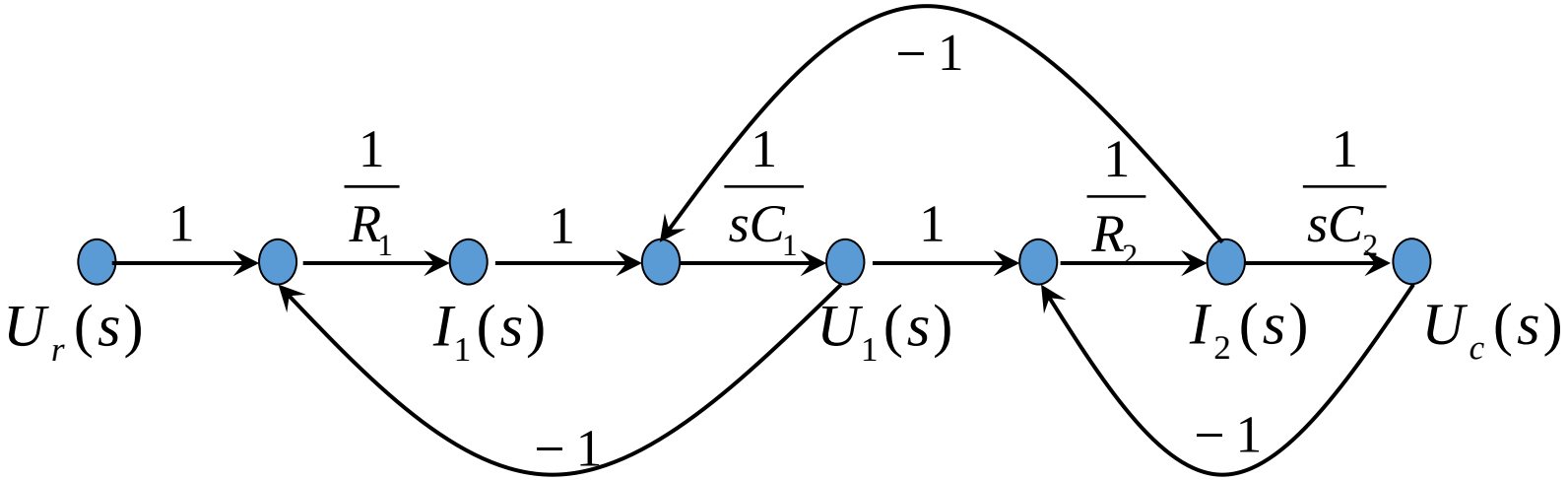
SFG was introduced by S. J. Mason for the cause-and-effect representation of linear systems.

1. Each signal is represented by a node.
2. Each transfer function is represented by a branch.



Block diagram and its equivalent signal-flow graph





Note

- A signal flow graph and a block diagram contain exactly the same information (there is no advantage to one over the other; there is only personal preference)

Mason's rule

$$M(s) = \frac{Y(s)}{U(s)} = \frac{1}{\Delta}\sum_{k=1}^{N} M_k \Delta_k$$

N = total number of forward paths between output $Y(s)$ and input $U(s)$

M_k = path gain of the k'th forward path.

Δ = $1 - \sum$(all individual loop gains)

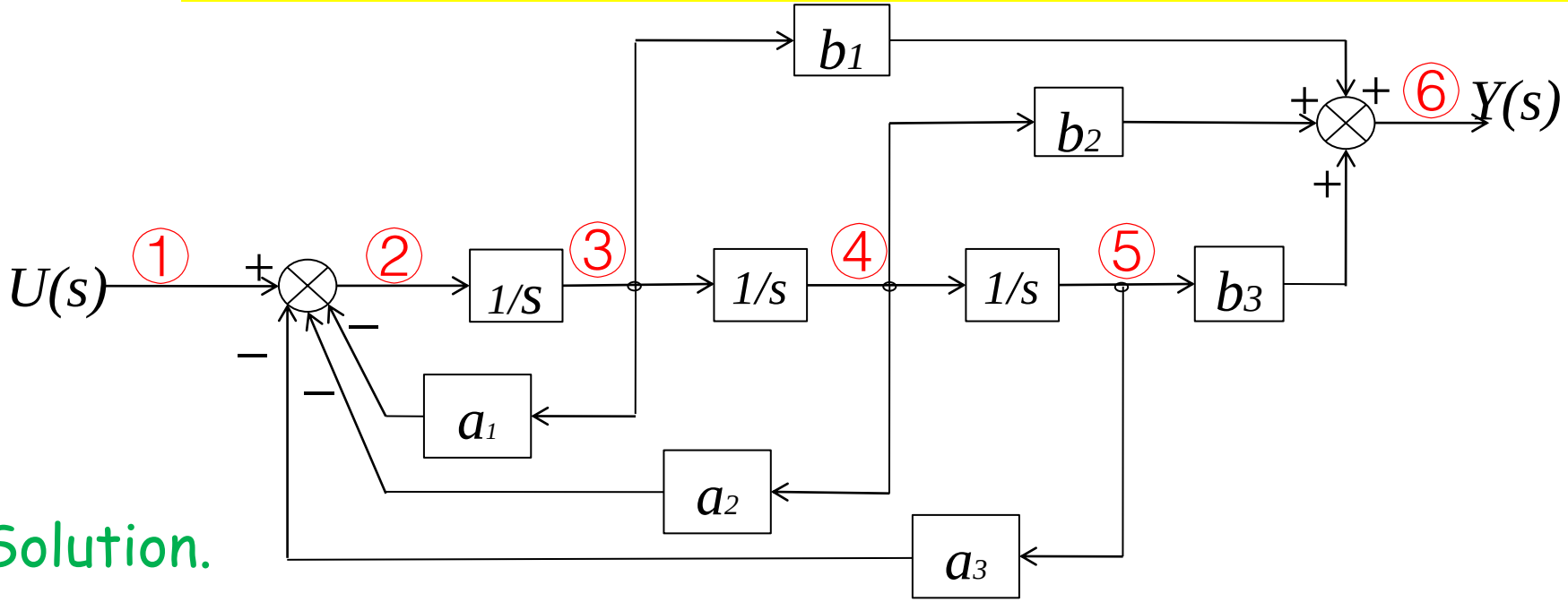
$+\sum$(gain products of all possible two loops that do not touch)

$-\sum$(gain products of all possible three loops that do not touch)

$+\cdots$

Δ_k = value of Δ for that part of the block diagram that does not touch the k'th forward path.

Example 1 **Find the transfer function for the following block diagram**



Solution.

Forward path Path gain

1236 $M_1 = 1\left(\frac{1}{s}\right)(b_1)(1)$

12346 $M_2 = 1\left(\frac{1}{s}\right)\left(\frac{1}{s}\right)(b_2)(1)$

123456 $M_3 = 1\left(\frac{1}{s}\right)\left(\frac{1}{s}\right)\left(\frac{1}{s}\right)(b_3)(1)$

and the Δ's are

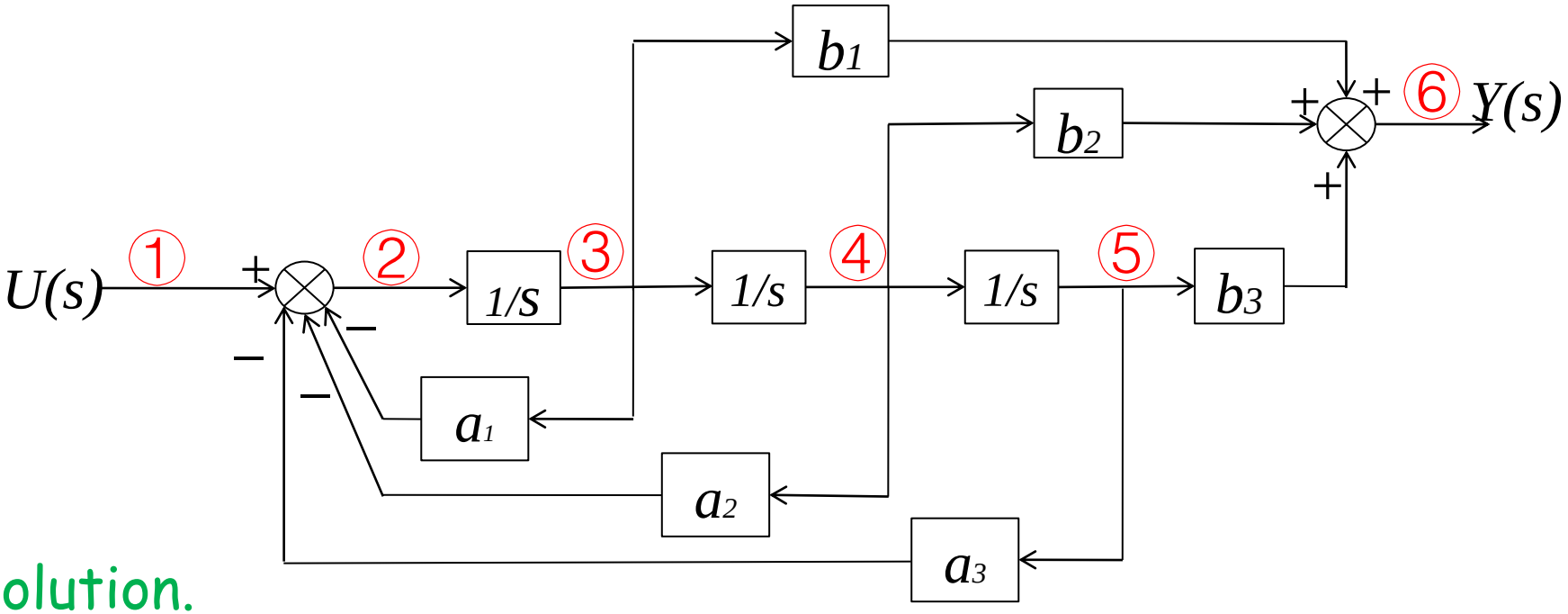
$$\Delta = 1 - \left(-\frac{a_1}{s} - \frac{a_2}{s^2} - \frac{a_3}{s^3}\right) + 0$$

$$\Delta_1 = 1 - 0$$

$$\Delta_2 = 1 - 0$$

$$\Delta_3 = 1 - 0$$

Example 1 **Find the transfer function for the following block diagram**



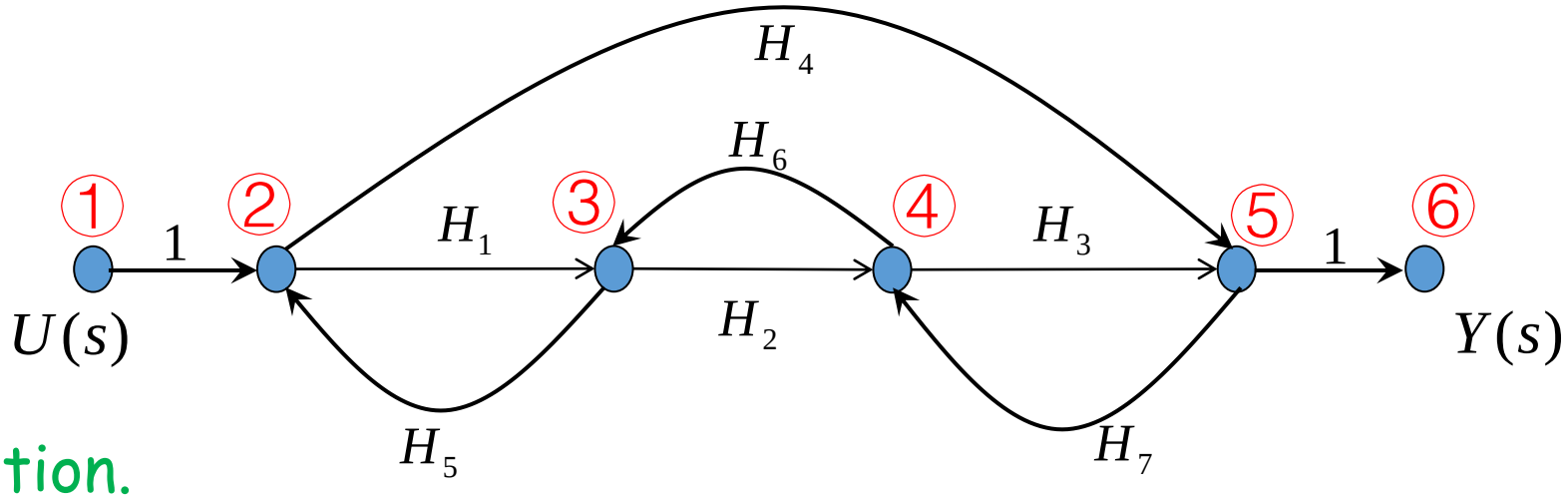
Solution.

Applying Mason's rule, we find the transfer function to be

$$M(s) = \frac{Y(s)}{U(s)} = \sum_{k=1}^{N} \frac{M_k \Delta_k}{\Delta}$$

$$= \frac{b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

Example 2 **Find the transfer function for the following SFG**



Solution.

Forward path	Path gain
123456	$M_1 = H_1H_2H_3$
1256	$M_2 = H_4$

Loop path	Path gain
232	$l_1 = H_1H_5$
343	$l_2 = H_2H_6$
454	$l_3 = H_3H_7$
25432	$l_4 = H_4H_7H_6H_5$

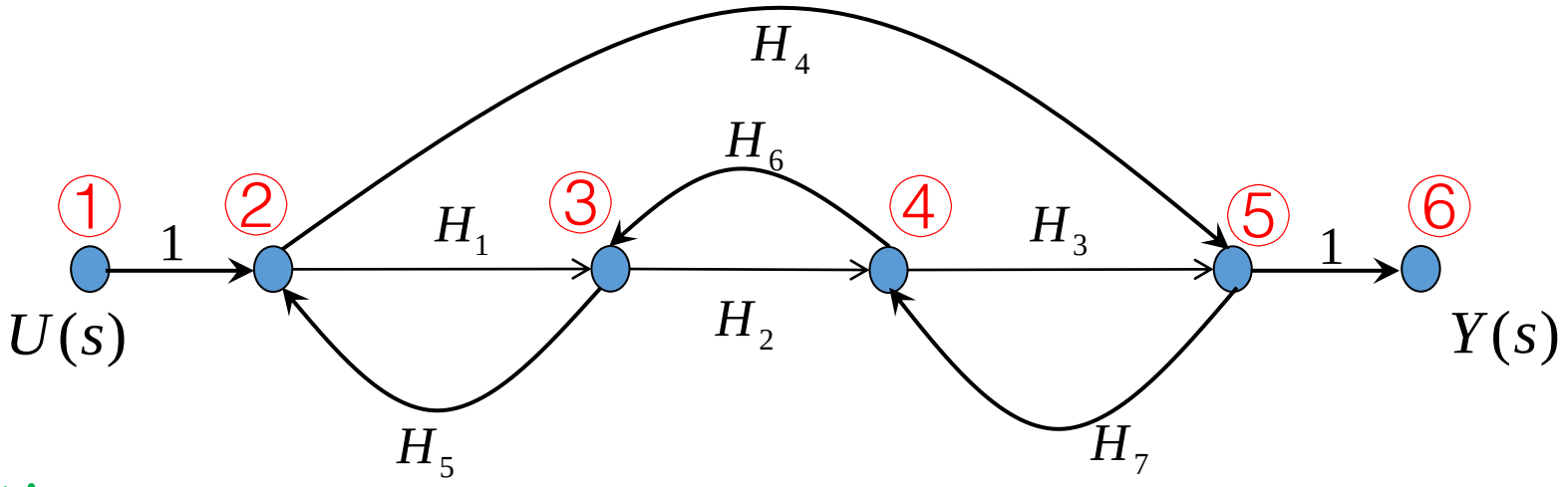
and the Δ's are

$$\Delta = 1-(l_1+l_2+l_3+l_4)+(l_1l_3)$$

$$\Delta_1 = 1-0$$

$$\Delta_2 = 1-H_2H_6$$

Example 2 Find the transfer function for the following SFG



Solution.

Applying Mason's rule, we find the transfer function to be

$$M(s)=\frac{Y(s)}{U(s)}=\sum_{k=1}^{N}\frac{M_k\Delta_k}{\Delta}$$

$$=\frac{H_1H_2H_3+H_4-H_4H_2H_6}{1-H_1H_5-H_2H_6-H_3H_7-H_4H_7H_6H_5+H_1H_5H_3H}$$